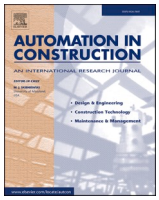
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UAV-based quantitative crack measurement for bridges integrating four-point laser metric calibration and mamba segmentation

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ABSTRACT

Crack width is an indicator of durability loss and serviceability in concrete bridges. Although UAV-based inspection is adopted, variable standoff distance and oblique imaging hinder valid, millimeter-level quantification. This paper presents a framework for crack identification and measurement. (1) A UAV-mounted four-point laser ranging device establishes a scale for each frame. Combined with homography and a Jacobian-based local length metric, the pixel-to-physical factor becomes a function of position and direction, which reduces scale drift across viewpoints. (2) CrackMamba-Net is designed to couple state space modeling with boundary sensitive representations, enhancing crack edge continuity and boundary clarity under fine and low contrast conditions. (3) Topology-preserving skeleton refinement with PCA-guided, distance-weighted linear correction estimates the local orientation; width is then measured along the refined normal and converted to physical units. Field and on-bridge experiments show linear agreement with references and low bias, supporting traceable, engineering-consistent crack quantification at the millimeter scale.

1. Introduction

Cracking is one of the most common and sensitive indicators of deterioration in concrete bridges, and its initiation and propagation are directly associated with reduced load-bearing capacity, degraded durability, and shortened service life [1]. Among various diagnostic metrics, crack width holds the most direct engineering significance. It is strongly linked to key degradation mechanisms such as moisture and chloride ingress and reinforcement corrosion [2], and also serves as a core threshold indicator in many international codes for condition assessment and maintenance decision-making [3–5]. Compared with simple crack presence detection, the ability to quantify crack width reliably and accurately under complex field conditions remains a central challenge in bridge health monitoring. Traditional manual inspection methods are limited by high subjectivity, low efficiency, and operational risk, making them inadequate for routine and fine-scale evaluation of large-scale infrastructure [6]. These limitations have driven increased research attention toward automated detection technologies focused on quantitative outcomes.

The UAV-mounted platform offers a highly maneuverable, low-interference, and cost-effective solution for rapid inspection and re-inspection of bridge surface defects. It enables high-resolution, multi-angle imaging of hard-to-access components such as webs, diaphragms, and box girder soffits, even under minimal or uninterrupted traffic conditions [7,8]. Kang et al. [9] proposed an autonomous structural health monitoring method that combines deep learning with an ultrasonic beacon system. By using geo-tagging, the approach enables precise defect localization, allowing UAVs to automatically identify and locate damage on bridge structures. Jiang et al. [10] developed a UAV system capable of switching between flight and climbing modes, enabling close-range acquisition of distortion-free crack images on tall towers and piers. Chen et al. [11] introduced an adaptive UAV-based bridge inspection strategy based on multi-level representation learning, which first performs coarse localization of structural components and then conducts fine-scale defect detection to balance efficiency and precision. Wang et al. [12] designed a UAV navigation method tailored for large-scale bridge environments without GPS signals. By integrating stereo visual-inertial odometry with fiducial marker correction, the method achieves accurate pose estimation to support bridge inspection tasks. Despite these advances, changes in standoff distance and imaging angle introduce perspective distortion and spatial scale inconsistency, which affect the reliability of geometric measurements [13]. Achieving

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millimeter-level precision in UAV-based quantification requires a reliable and transferable scale factor that remains geometrically consistent across positions and orientations within the image. Ensuring such consistency across the field of view remains a key limitation in existing approaches [14].

To ensure the feasibility and reliability of quantitative analysis, it is essential to first achieve accurate crack boundary identification and segmentation. This provides geometric constraints that support the subsequent measurement process. In recent years, deep learning techniques have rapidly advanced in the fields of image recognition and structural defect detection [15,16], With capabilities such as end-to-end training, autonomous feature extraction, and model adaptability, they have become the mainstream approach for intelligent recognition of surface-level defects [17,18]. Pixel-level segmentation models, in particular, enable fine-grained delineation of target boundaries and have been widely applied in concrete crack detection tasks. Liu et al. [19] were the first to introduce the U-Net model for crack detection in concrete. This pioneering work demonstrated high efficiency and robustness in localizing crack regions within input images. In related studies, Chen et al. [20] proposed an attention-based online monitoring framework, employing a lightweight network for crack segmentation. To enhance the capture of crack details, Ong et al. [21] developed a feature pyramid network embedded with a self-guided attention module. This architecture was specifically designed to address the complexity of crack segmentation scenarios, enabling more precise and enhanced representation of fine crack features. Despite their strong capability in local modeling, convolution-based segmentation frameworks are inherently limited in capturing long-range dependencies across an image, which constrains their ability to fully perceive crack morphology and spatial distribution. With the introduction of the Transformer architecture, self-attention mechanisms have been incorporated into vision models, significantly enhancing their ability to model global semantic dependencies [22,23]. Kang et al. [24] developed a non-local micro-crack segmentation network based on the Transformer, which integrates global contextual information through non-local modules and achieves efficient segmentation of fine crack boundaries. He et al. [25] proposed the BGCrack model, which combines boundary information with global perception modules to further improve pixel-level segmentation accuracy. Although Transformers excel at modeling long-range dependencies, their high computational cost when processing high-resolution images has become a major bottleneck limiting practical deployment. In response, the recently proposed Mamba architecture, built upon state space models (SSMs), retains the capacity for global dependency modeling while significantly reducing computational complexity [26]. Through sparsification and a near-linear computational path, Mamba efficiently captures long-range correlations and offers a better trade-off between computational efficiency and representational power, making it a promising alternative to Transformers. Several studies have explored the integration of Mamba into crack segmentation tasks. For instance, Tang et al. [27] embedded Mamba modules into the U-Net framework to enhance the stability of crack contour extraction. Zuo et al. [28] introduced a Mamba-based segmentation model, CrackMamba, which incorporates a Snake Scan sequencing strategy and a three-branch SCVSS module to fuse local and global features, enabling efficient and accurate segmentation of structurally complex cracks. Han et al. [29] combined residual visual Mamba blocks with convolutional neural networks to improve sensitivity to fine cracks and suppress background interference. Zhang et al. [30] further enhanced micro-crack detection by introducing a frequency-domain feature fusion strategy and cross-visual Mamba branches, which jointly strengthen crack responses and suppress noise. As a novel state space modeling paradigm, Mamba has demonstrated strong robustness and scalability in concrete crack segmentation. Leveraging its balance between global modeling capability and computational efficiency, this study adopts Mamba modules to construct a segmentation framework aimed at achieving high-precision, fine-grained representation of surface cracks in concrete structures.

Segmentation provides pixel-level morphological priors and topological constraints for the crack region, yet it only marks the beginning of the quantification pipeline. The engineering definition of crack width is explicit: it is the shortest physical distance between the two crack boundaries measured along the local normal direction at the crack centerline [31]. Under this geometric definition, the core task is to couple accurate determination of the local normal with position-dependent pixel-to-physical scale recovery, so that directional selection and scale measurement remain consistent within a common coordinate frame. Lins et al. [32] proposed an automated crack detection and geometric measurement system based on single-camera image analysis, aiming to replace traditional manual inspection processes and significantly improve detection efficiency and measurement accuracy. Kouzehgar et al. [33] introduced a self-reconfigurable exterior wall cleaning robot, equipped with an online crack detection system based on convolutional neural networks, enabling the robot to automatically avoid crack regions in high-risk, multi-glass curtain wall environments. Alamdari et al. [34] explored a multi-scale robotic inspection system that combines deep learning-based visual recognition with high-precision scanning for the automated measurement of microcrack width, length, and geometry in concrete structures. Kim et al. [35] proposed an end-to-end framework based on Building Information Modeling (BIM) and 3D laser scanning technology for the automated evaluation of the dimensions and surface quality of precast concrete components, aiming to replace traditional manual measurement methods. Kang et al. [36] designed a hybrid approach that combines deep learning with image processing to achieve pixel-accurate crack segmentation and geometric quantification under complex backgrounds. Ong et al. [37] proposed a hybrid algorithm that merges the shortest-distance method with orthogonal projection, selecting nearest point pairs along the crack normal direction to improve measurement accuracy across diverse geometries. Li et al. [38] developed the Ortho-Boundary algorithm, which uses crack skeleton orientation and boundary orientation to generate measurement directions and enables rapid, accurate width estimation. Kim et al. [39] used UAV imagery within a joint mapping based on known bridge component dimensions to estimate crack width and length. Li et al. [40] established a calibrated mapping between pixel area and true area to quantify multiple types of concrete defects at the pixel level, thereby converting segmentation outputs into actual geometric indicators. Ding et al. [41] proposed a markerless field scale calibration method that indexes pixel scale factors across different camera poses and converts pixel measurements to real-world dimensions for quantitative analysis. Peng et al. [42] built a pixel scale factor and ranging calibration model through a hybrid feature learning strategy, enabling accurate conversion from pixel measurements to actual crack width in UAV images. Germanese et al. [43] attached planar markers near cracks and derived the pixel-to-physical scale factor from their geometry in UAV images, thereby mapping measurements from pixel space to real-world size. Although current crack width estimation workflows are usable, key deficiencies remain. First, from a geometric definition perspective, crack shapes typically exhibit non-linear features with bends and turns. If local orientation is neglected during width measurement and a fixed direction or simplified approximation is used, the measurement is prone to significant structural errors due to deviation from the correct geometric definition. Second, in terms of physical scale acquisition, traditional calibration methods heavily rely on manually placing reference markers, such as attaching reference points or objects near the crack. This contact-based approach is not only inefficient but also impractical for large-scale bridge inspections, undermining the core advantage of UAVs for remote, non-contact operations. Third, in automated estimation, some methods attempt to estimate scale factors based on theoretical ground sampling distance (GSD) or fixed distance assumptions. However, UAVs inevitably experience variable standoff distances and oblique imaging due to dynamic airflow, causing real-time fluctuations in GSD. This leads

to significant systematic bias and perspective distortion when relying on a single global scale factor, resulting in errors across both spatial and temporal dimensions.

To the best of our knowledge, no existing study has yet integrated dynamic scale calibration, fine-grained semantic segmentation, and geometrically rigorous crack width measurement into a fully closed-loop UAV workflow. To fill this gap and effectively address the scale uncertainty introduced by UAV variable standoff distances and oblique imaging, as well as the geometric deviations inherent in traditional crack width measurement methods, the core contribution of this study lies in the injection of physical scale into each image frame using a four-point laser reference. This approach enables the accurate conversion of pixel-based crack width to a true physical measurement. Building on this, a comprehensive technical framework is proposed to seamlessly integrate UAV-mounted four-point laser external scale injection with imaging geometry modeling. The method employs the UAV-mounted four-point laser ranging device as an external absolute scale source. Based on distance observations along four known beam directions and the inter-beam angles, an absolute four-point reference geometry is reconstructed on the target plane. The pixel coordinates of the four red laser spots are then robustly extracted using zero-mean normalized cross-correlation within a fixed region of interest. The normalized DLT is then applied to estimate the homography matrix, and the derivative of the induced mapping is used to obtain the Jacobian matrix. This allows the definition of a position- and direction-dependent local length metric (mm/px) at any pixel location, creating a geometry-consistent mapping from pixel to physical scale. For the crack segmentation task, CrackMamba-Net, which integrates Mamba state space modeling, is employed. Unlike general segmentation models that primarily focus on pixel classification accuracy, this network specifically emphasizes the continuity of crack topology and boundary integrity. Leveraging the advantages of the Mamba state space model in long-sequence modeling, it effectively addresses issues with crack discontinuity in fine cracks under complex textures, producing high-fidelity continuous masks. This feature is a prerequisite for subsequent skeleton extraction and high-precision geometric quantification, clearly distinguishing it from purely visual segmentation approaches. In the quantification stage, the method performs skeleton refinement followed by principal component analysis and distance-weighted linear correction to estimate the local dominant direction. Intersections between the refined normal and both crack boundaries are used to obtain the pixel-domain width, which is then converted into physical length through the local length metric. The proposed framework achieves a closed-loop coupling of absolute scale injection, projective geometry mapping, and local metric computation within the same frame. It effectively suppresses scale drift caused by variations in viewing distance and angle, while ensuring that the measured width remains consistent with the actual crack orientation. This provides a systematic and traceable approach for millimeter-level crack width quantification under UAV-mounted scenarios.

The paper is organized as follows. Section 2 presents the overall research methodology. Section 3 describes the UAV-mounted four-point laser ranging device and the pixel-to-physical scale mapping method in detail. Section 4 provides the implementation details of the crack segmentation network (CrackMamba-Net). Section 5 introduces the algorithmic design for crack width measurement. Section 6 discusses the execution details and outcomes of the field experiments. Finally, Section 7 summarizes the main contributions of this study and outlines directions for future research.

2. Overall framework and technical pipeline

To overcome the limitations of traditional methods that are restricted to pixel-scale estimation or heavily reliant on manually preset reference objects, and to address the challenges UAVs face in obtaining reliable absolute scale and geometrically accurate crack width measurements under oblique imaging and variable standoff conditions, this study proposes an end-to-end closed-loop method chain based on UAV-mounted four-point laser auto-calibration (overall workflow shown in Fig. 1). The framework eliminates the reliance on fixed shooting distances or contact-based markers by actively projecting a laser array to achieve non-contact physical scale injection. This is combined with high-fidelity visual segmentation and geometric measurement, creating a fully automated closed-loop from “absolute scale recovery” to “crack semantic extraction” and “millimeter-scale width quantification” within a single frame.

The framework first establishes the scale injection mechanism, using a four-point laser ranging device as the external absolute scale source. Based on distance observations and inter-beam angles, a planar reference geometry is reconstructed, and pixel coordinates of the laser spots are simultaneously extracted. The normalized DLT is applied to estimate the homography matrix, and its derivative is used to obtain the Jacobian matrix, thereby establishing a position- and direction-dependent local length metric, which is geometrically consistent with the imaging geometry. Building on this scale reference, CrackMamba-Net integrates Mamba state space modeling to provide high-fidelity visual perception. This network works by combining a global–local visual state space module (GL-VSS) at the encoder and a multi-scale convolution module (MSC-VSS) at the decoder, ensuring global topological consistency while precisely capturing the edges of fine cracks. Finally, the process transitions to geometric quantification, where the skeleton of the segmented mask is used to estimate the dominant direction through PCA within the local neighborhood. A distance-weighted first-order linear fit refines the direction, and the intersections with the crack boundaries along the refined normal direction are used to obtain pixel width. This is then converted to physical units using the local length metric generated in the first stage.

By integrating external absolute scale injection, projective geometry, and local metric coupling in a single frame, this framework effectively suppresses scale drift caused by viewing angle and flight distance variations and ensures that crack width definition is strictly aligned with the crack orientation. As a result, it enables millimeter-level, traceable crack quantification for UAV-based inspection scenarios.

3. Image scale calibration based on a four-point laser ranging device

To address the significant positional variation of image scale under oblique imaging and variable flight distance in UAV-mounted scenarios, where a single ground sampling distance cannot support millimeter-level quantification, this study develops a hardware-software integrated scale calibration framework. The core concept is to employ a four-point laser ranging device as an external absolute scale source. With a lightweight carbon-fiber body and an adjustable-angle gimbal, the system maintains high geometric observability across varying flight distances and viewing angles. Synchronized with the camera exposure through hardware triggering, the device outputs four distance measurements and beam direction angles at the same instant. Based on these data, the true side lengths among the four points are reconstructed on the measured plane, forming a two-dimensional physical reference geometry with absolute units. In the image processing stage, zero-mean normalized cross-correlation is applied within a fixed region of interest to robustly extract the pixel coordinates of the four laser spots. A one-to-one correspondence is then established between these pixel coordinates and the physical reference geometry. The normalized Direct Linear Transformation is used to estimate the homography mapping from the image plane to the physical plane, and the Jacobian matrix derived from this mapping defines a position-dependent local length metric in millimeters per pixel, which avoids resampling of the entire image. This framework injects an external absolute scale through coaxial laser measurements, establishes a geometric mapping from pixel space to the physical plane using the homography relationship, and derives a spatially adaptive metric through the Jacobian matrix. As a

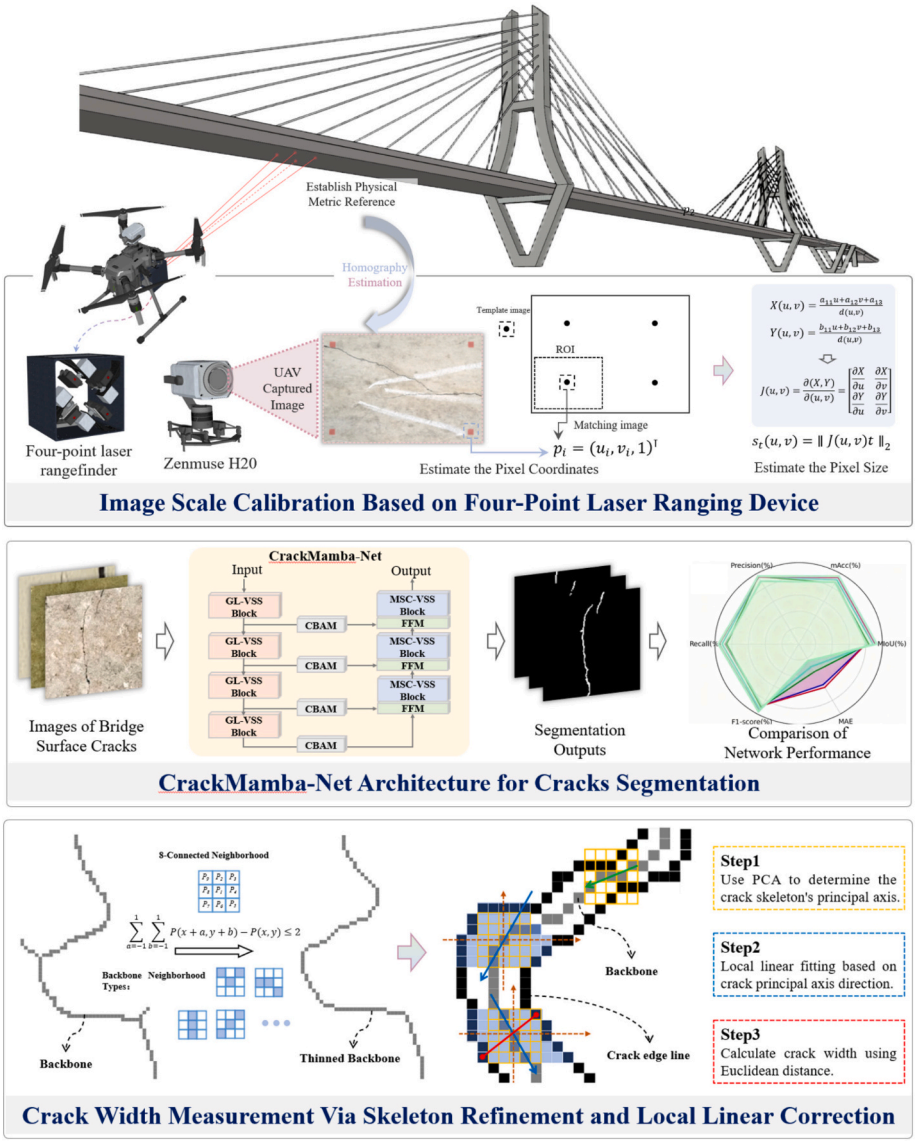


Fig. 1. Framework of the proposed UAV-assisted crack detection and width quantification method.

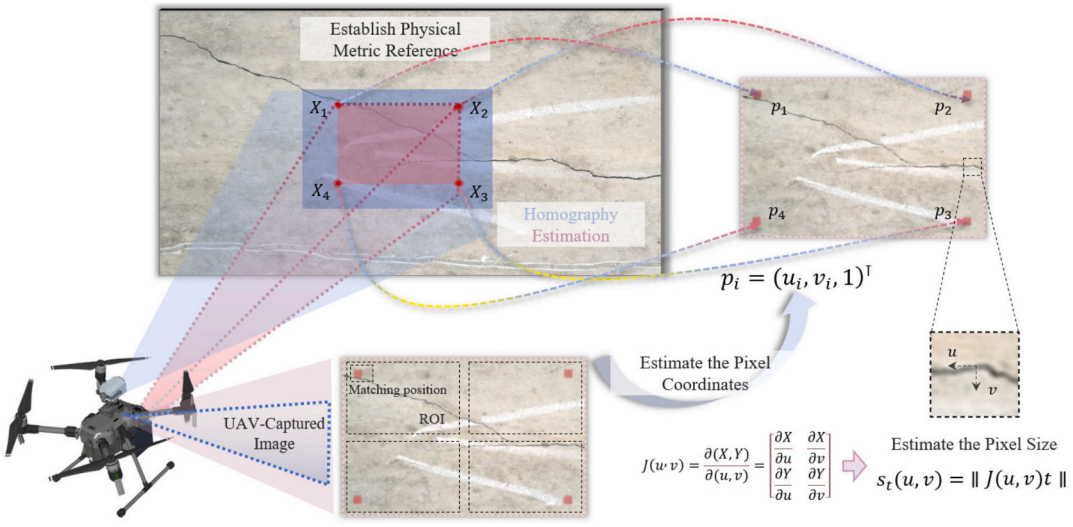


Fig. 2. UAV-based image scale calibration using a four-point laser ranging device.

result, raw pixel measurements are converted into physical quantities in a stable manner, which provides reliable scale support for high-precision quantification of local geometric parameters such as crack width. The overall process is illustrated in Fig. 2.

3.1. System design and implementation of the four-point laser ranging device

The design core of the four-point laser ranging device lies in establishing a traceable scale reference through an adjustable geometric baseline combined with a highly stable mechanical structure. The system arranges four laser ranging sensors symmetrically along the diagonals at 90 degrees and mounts them on a dedicated bracket fixed to the stepper motor shaft, generating a rectangular laser spot array on the target plane. As the motor continuously rotates, the side length of the array becomes adjustable from small to large, ensuring suitable scale coverage and enabling integrated calibration across different working distances. Specifically, under long-distance or oblique imaging conditions, the inter-beam angle is increased to enlarge the spacing of the four points on the physical plane, thereby expanding the geometric scale, improving the projection condition number, and enhancing the robustness of the homography estimation. In contrast, for close-range operations or partially occluded scenes, the inter-beam angle is reduced to ensure that all four laser spots remain within the camera's field of view. It should be specifically noted that adjustments to this geometric configuration are task-level settings designed for specific inspection conditions, such as switching from a global overview to a localized close-up inspection. This operation is typically performed before flight or during hover gaps through remote commands. Once the angle is set, the stepper motor remains locked during the subsequent continuous data collection sequence, rather than undergoing real-time dynamic adjustments during each frame exposure. Consequently, the scale constraint no longer depends on external reference boards or additional baseline calibration. Instead, it is directly determined by the device's intrinsic geometric configuration and ranging measurements, providing a stable and reliable absolute reference for the subsequent conversion from pixel scale to physical scale.

The device adopts a modular system architecture composed of laser ranging sensors, a programmable logic controller (PLC), a stepper motor with its driving module, and human-machine interaction and remote control units, as illustrated in Fig. 3. To balance lightweight design and structural rigidity, the housing is made of carbon-fiber composite material, which provides high stiffness and vibration resistance at low density. This design minimizes the effect of attitude disturbance on measurement stability while meeting the UAV-mounted platform's payload constraints. The measurement channel employs a commercial Myantenna L2s-80 laser ranging sensor as the core of each ranging unit. The sensor covers a measurement range of 0.03 m to 80 m with an absolute accuracy of ±1 mm. Data are transmitted via RS-485 and are compatible with the Modbus RTU protocol, enabling seamless communication with both the control and upper-level systems. This sensor achieves a good balance between range, precision, and power consumption, ensuring the feasibility of UAV-mounted integration and providing stable metrological performance for long-term continuous operation. The control unit is based on the XINJE XL3 series PLC, which, with its sufficient I/O resources and reliable high-speed processing capacity, performs real-time acquisition, centralized scheduling, and anomaly monitoring of four-channel laser ranging data and motor status. This ensures time synchronization and data consistency at the system level. The actuator adopts a two-phase hybrid stepper motor, and the paired driver supports step modes from full-step to 1/32 microstepping. Even under low power constraints, it enables high-resolution and low-vibration angular adjustment, ensuring precise and repeatable control of the laser spot array side length. To enhance operability and remote maintenance, the system integrates a Haiwell touch panel and the Haiwell Cloud App. The touch panel provides local task configuration and data visualization, while the cloud application enables secure remote access and parameter delivery for mobile devices, allowing operators to perform real-time monitoring and strategy adjustment via smartphone. Overall, the proposed system achieves an effective balance among lightweight design, modular configuration, and remote controllability. It meets the payload and energy limitations of UAV-mounted platforms and continuously provides stable, synchronized, and high-quality input data for subsequent algorithmic processing.

The system operates in three stages: initialization and calibration, real-time measurement, and remote control. During the initialization stage, the PLC drives the stepper motor according to preset angles to rotate the four laser ranging sensors into the target orientation. This forms a rectangular laser spot array with a predefined side length, establishing the initial geometric scale and storing the corresponding angular parameters. This process provides a traceable scale reference for subsequent operations. It is worth noting that, although the device's housing is made from high-stiffness carbon fiber composite material to resist mechanical deformation, long-term micro-movement wear of the

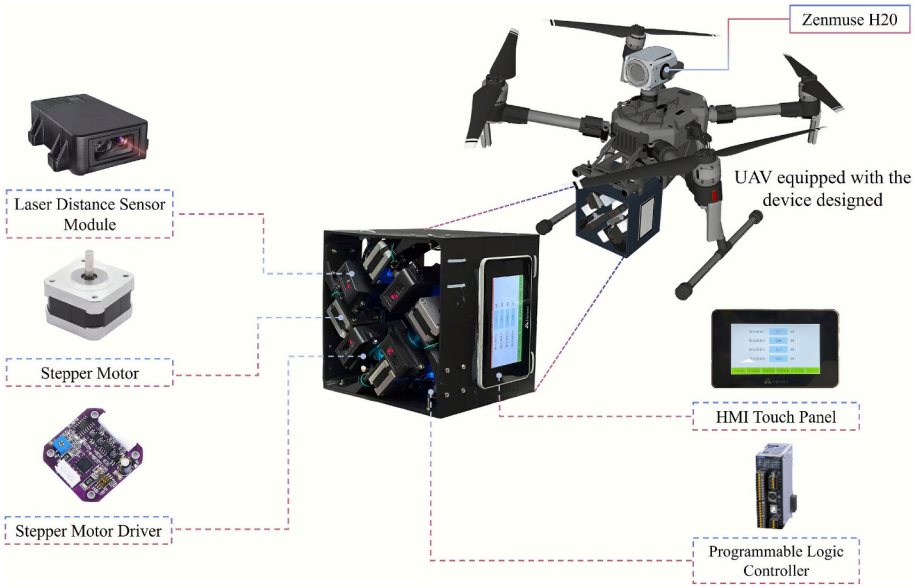


Fig. 3. Hardware configuration of the UAV-mounted four-point laser ranging device.

motors may still cause angular deviation. To address this issue, the pre-flight initialization process designed in this study effectively mitigates this problem by forcibly resetting the motors to the hard limit and regenerating the preset rectangular geometry before each operation. This step not only establishes an immediate scale reference but also serves as a calibration to eliminate accumulated mechanical drift, ensuring the geometric stability of the measurement system during long-term use. Once the measurement phase begins, the laser ranging sensors output four-channel distance data at 20 Hz (configurable to 10 Hz to match the UAV camera frame rate and trigger timing, depending on the task requirements). The controller timestamps the distance readings and motor angles uniformly, and a hard-wired trigger signal from the PLC output is synchronized to the control unit to drive camera exposure and distance sampling. Considering the signal transmission delay (in microseconds) and internal integration time of the sensors, the system achieves millisecond-level time alignment accuracy, with effective synchronization error controlled within 20 ms. To analyze potential scale deviations introduced during synchronization, an error sensitivity analysis was conducted based on the UAV's hover operation (with flight speed $v < 0.1$ m/s). The maximum displacement error in the view axis due to synchronization delay is approximately 2 mm, which, relative to the operational distance of 2–6 m, results in a relative scale error of less than 0.1%, well below the system's overall measurement uncertainty. This mechanism ensures strict spatiotemporal pairing between each image frame and the corresponding four laser distance measurements, thus guaranteeing the geometric rigor of the pixel-to-physical mapping under dynamic data collection conditions. Locally, the Haiwell touch panel enables visualization and configuration of the four-channel distance data, as shown in Fig. 4(a). Since the Euclidean relationships of the laser spots on the measured plane are fully determined by the distance outputs and the known inter-beam angles, each synchronized data pair provides an absolute scale input that directly supports the conversion from pixel coordinates to physical distances and enables robust estimation of the homography mapping. In the remote control stage, the Haiwell touch panel continuously displays the ranging curves and system status while automatically recording historical data for post-analysis and system health diagnosis. Meanwhile, the Haiwell Cloud App provides cross-platform access, allowing researchers to adjust parameters and update control strategies in real time during UAV operation. This ensures consistent scale constraints under varying viewing distances and imaging angles, as illustrated in Fig. 4(b). Through this workflow, the system establishes a closed-loop mechanism integrating hardware control, geometric calibration, data acquisition, and remote management. This integration ensures both engineering practicality and metrological consistency, enabling accurate and repeatable millimeter-level scale recovery for UAV-mounted crack width measurement.

3.2. Construction of the physical reference and determination of laser spot pixel coordinates

The proposed hardware configuration ensures that the four laser beams and the UAV-mounted camera are both temporally synchronized and geometrically measurable. Based on the output distance and directional information, the true side lengths among the four laser spots on the target plane can be reconstructed, allowing the establishment of a two-dimensional physical coordinate system $\{X_i\}$ with absolute units. This serves as a unified geometric reference for image measurements under different poses and working distances. This method is based on a key geometric assumption: the crack surface within the laser projection area is considered a local plane. Given that the UAV's close-range detection typically covers a small field of view (usually ranging from 0.5 m to 1 m), and the primary components of concrete bridges, such as box girder soffits and pier webs, have very low curvature at the local scale, this assumption is widely applicable in engineering practice.

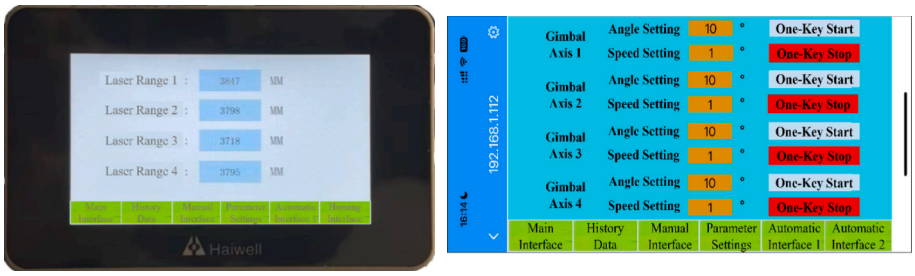
Let the four co-axial laser beams form spot centers $P_i (i = 1, \ldots, 4)$ on the measured plane, with effective path lengths r_i along their respective directions and inter-beam angles θ_{ij}. The optical emission center of the device is denoted as O, and the fixed mechanical offsets h_i between O and each ranging channel are determined during calibration. The effective distance used in this method is $r_i = l_i + h_i$, where l_i is the raw measurement from the laser ranging sensor. According to the cosine law, the Euclidean distance between two laser spots is

$$d_{ij} = \| P_i - P_j \| = \sqrt{r_i^2 + r_j^2 - 2 r_i r_j \cos\theta_{ij}} \tag{1}$$

and the set $\{d_{ij}\}$ (unit: mm) provides the geometric constraints of the measurement plane, as shown in Fig. 5(a). A two-dimensional physical coordinate system $(X, Y)$is then constructed on the plane. Without loss of generality, the first laser spot is defined as the origin $X_1 = (0, 0, 1)^\intercal$, and the second laser spot is placed on the x-axis as $X_2 = (W, 0, 1)^\intercal$, where $W = d_{12}$. The coordinates of the fourth laser spot X_4 are determined using the intersection of two circles, as shown in Fig. 5(b). Given $\| X_4 - X_1 \| = d_{14}$ and $\| X_4 - X_2 \| = d_{24}$, two circles are constructed with centers at X_1 and X_2 and radii d_{14} and d_{24}, respectively. The intersection points of these circles provide the coordinates of X_4. The specific calculation is as follows:

$$x_4 = \frac{W^2 + d_{14}^2 - d_{24}^2}{2W}, y_4 = +\sqrt{d_{14}^2 - x_4^2} \tag{2}$$

where the positive sign fixes the plane orientation, yielding $X_4 = (x_4, y_4, 1)^\intercal$. As shown in Fig. 5(c), the position of X_3 is determined using geometric relationships. Given that $X_4 = (x_4, y_4, 1)^\intercal$ is derived from the known distances d_{14} and d_{24}, and $X_2 = (W, 0, 1)^\intercal$ is known, the distance $D = \| X_4 - X_2 \| = d_{24}$ is defined. Next, the direction vector $\mathbf{e}_x = \frac{X_4 - X_2}{D}$ and normal vector $\mathbf{e}_y = (-e_{x,y}, e_{x,x})$ are calculated, where $\mathbf{e}_x$ and $\mathbf{e}_y$ form an orthogonal right-hand coordinate system with unit vectors. Using the



(a) Four-channel range visualization on the Haiwell HMI.

(b) Mobile client for remote control: gimbal angle/speed settings and start-stop commands.

Fig. 4. HMI & Mobile remote control interfaces of the four-point laser ranging device.

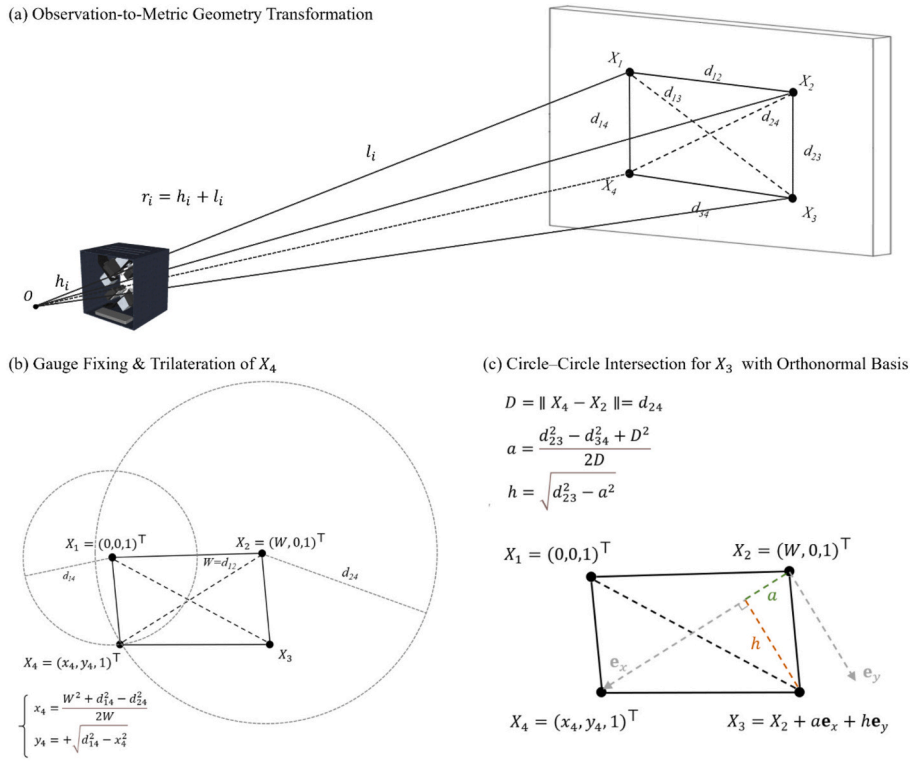


Fig. 5. From four-point laser ranging observations to planar metric embedding.

geometric relationship, the projection along $\mathbf{e}_x$ and the normal offset h are calculated as follows:

$$a = \frac{d_{23}^2 - d_{34}^2 + D^2}{2D}, h = \sqrt{d_{23}^2 - a^2} \tag{3}$$

Finally, the coordinates of X_3 are obtained by: $X_3 = X_2 + a\mathbf{e}_x + h\mathbf{e}_y$. This calculation defines the coordinates of X_3 based on the projections and offsets along the directions of $\mathbf{e}_x$ and $\mathbf{e}_y$, starting from X_2. Thus, the four reference points on the measured plane have true two-dimensional coordinates $\{X_i = (x_i, y_i, 1)^\top\}_{i=1}^4$, whose internal Euclidean geometry is strictly consistent. This coordinate system therefore possesses explicit physical scale and metrological significance. The constructed physical reference serves as the physical-plane point set in the subsequent homography estimation, corresponding one-to-one with the four image-domain pixel observations $\{p_i\}$. It defines the pixel-to-physical mapping and anchors the conversion of crack width from pixels to millimeters.

To extract the pixel coordinates of the four laser spots from UAV-acquired images and ensure stable performance under oblique imaging and variable distance conditions with minimal computational complexity, this study employs a fixed region of interest (ROI)-based template correlation method for pixel-level localization. Specifically, the process begins with geometric priors by dividing the image into four fixed ROIs corresponding to the typical landing areas of the four laser beams. Considering the spectral characteristics of the red laser, detection is performed on an enhanced red channel defined as $R^* = R - max(G, B)$, which suppresses non-target reflections and background color interference. Each ROI undergoes mild denoising and morphological filtering to remove slow-varying backgrounds and isolated bright noise. Within each ROI, the algorithm first identifies the brightest pixel as a candidate laser spot center and extracts a small local template centered on it. To reduce the influence of illumination changes and local contrast variation, the template is normalized to zero mean and unit variance, achieving robustness against brightness fluctuations. Subsequently, Zero-mean Normalized Cross-Correlation (ZNCC) is used to compute the correlation response surface between the template and the image region. As a commonly used structural similarity measure, ZNCC effectively compensates for global and local intensity shifts within small windows, thereby ensuring detection accuracy. Let the template support domain be Ω, and the template and image patches be $T(x, y)$ and $I(x, y)$, respectively. When the template is translated to position (u, v), ZNCC is defined as

$$\mathrm{ZNCC}(u,v) = \frac{\sum_{(x,y)\in\Omega} \left(I(u+x, v+y) - \bar{I}\right)\left(T(x,y) - \bar{T}\right)}{\sqrt{\sum_{(x,y)\in\Omega} \left(I(u+x, v+y) - \bar{I}\right)^2} \sqrt{\sum_{(x,y)\in\Omega} \left(T(x,y) - \bar{T}\right)^2}} \tag{4}$$

where $\bar{I}$ and $\bar{T}$ are the local and template means, respectively. The integer-grid position of the correlation peak, denoted as (i_0, j_0), represents the pixel coordinates of the laser spot, as illustrated in Fig. 6. To quantify detection confidence, the peak distinctiveness (the difference or ratio between the primary peak and its nearest secondary peak) is recorded. When this value falls below an empirical threshold (0.05–0.1), the search window of the ROI is expanded or the threshold relaxed for recalculation, thereby reducing the risk of false peak detection.

3.3. Derivation of the pixel-to-physical scale factor based on Homography

After constructing the physical reference of the measured plane and accurately extracting the laser spot pixel coordinates, strict Lens Distortion Correction must be performed. As shown in Fig. 7, real optical imaging systems inevitably exhibit radial distortion (manifesting as barrel or pincushion distortion) and tangential distortion. These nonlinear errors contradict the ideal pinhole model assumption upon which homography transformations rely, and if uncorrected, would lead to significant errors in the scale measurement derived from the Jacobian matrix. To address this, Zhang's Camera Calibration method [44] is used to obtain the distortion parameters (including radial coefficients k and tangential and decentering distortion coefficients p, s), which are then

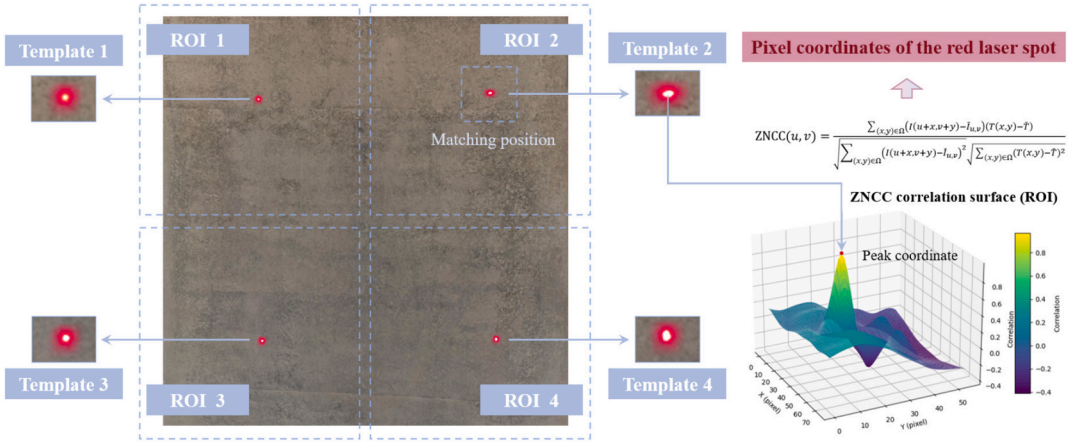


Fig. 6. ROI-partitioned ZNCC template matching for pixel localization of four red laser spots. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)



Fig. 7. Integrated pipeline of lens distortion correction.

applied to undistort the raw pixel observations. This restores the imaging geometry to its ideal linear projection state. Based on this, homography transformation is used to map the geometrically corrected pixel coordinates to a physical coordinate system with absolute scale, effectively eliminating the perspective effects introduced by variations in camera pose and distance, and establishing a unified geometric channel for high-precision pixel-level quantification. Let the homogeneous pixel coordinates of the four laser spots in the image domain be $p_i = (u_i, v_i, 1)^{\mathrm{T}}$, and their corresponding homogeneous coordinates in the physical reference plane be $X_i = (x_i, y_i, 1)^{\mathrm{T}}$. Under the pinhole camera model, any point pair on the same plane satisfies a homography relationship. Therefore, there exists a matrix $H \in \mathbb{R}^{3\times3}$ such that

$$s_i X_i = H p_i (i = 1, \ldots, 4) \tag{5}$$

where $s_i \neq 0$ is the homogeneous scale factor. A direct linear solution is numerically sensitive to coordinate magnitudes and point distributions. To improve robustness, this study adopts the Normalized Direct Linear Transformation. Specifically, both point sets $\{p_i\}$ and $\{X_i\}$ are translated and uniformly scaled so that their centroid is at the origin and their mean distance equals $\sqrt{2}$. The homography matrix $\widehat{H}$ is then estimated in the normalized space using Singular Value Decomposition (SVD), followed by denormalization to obtain H:

$$H = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ b_{11} & b_{12} & b_{13} \\ \gamma_1 & \gamma_2 & \gamma_3 \end{bmatrix} \tag{6}$$

After obtaining the homography mapping from the pixel domain to the physical plane, it is necessary to convert this global mapping into a local metric relationship, that is, to map an infinitesimal pixel displacement at any image position to its corresponding real-world length on the measured plane. This differential metric is required because perspective projection causes the scale to vary with position; the same pixel step corresponds to different physical lengths at the image center and edges. Continuing to use a single global scale factor would inevitably introduce systematic bias near image boundaries or under oblique viewing and variable distance conditions. To address this, the homography mapping is differentiated with respect to pixel coordinates (u, v), yielding the Jacobian matrix that relates pixel differentials to physical differentials. This provides a precise millimeters-per-pixel metric at every point and in every direction. The homography can thus be expressed in its rational form as follows:

$$\begin{aligned} X(u,v) &= \frac{a_{11}u + a_{12}v + a_{13}}{d(u,v)}, Y(u,v) = \frac{b_{11}u + b_{12}v + b_{13}}{d(u,v)}, d(u,v) \\ &= \gamma_1 u + \gamma_2 v + \gamma_3 \end{aligned} \tag{7}$$

Here (X, Y) are measured in millimeters and (u, v) in pixels. Since the homography is a smooth projective mapping within the valid imaging region, the function is differentiable in this domain, and its first-order partial derivatives form the Jacobian matrix:

$$J(u,v) = \frac{\partial(X,Y)}{\partial(u,v)} = \begin{bmatrix} \frac{\partial X}{\partial u} & \frac{\partial X}{\partial v} \\ \frac{\partial Y}{\partial u} & \frac{\partial Y}{\partial v} \end{bmatrix} \tag{8}$$

By applying the differentiation rule for rational functions, a closed-form expression can be derived:

$$\frac{\partial X}{\partial u} = \frac{a_{11}d - X\gamma_1}{d^2}, \frac{\partial X}{\partial v} = \frac{a_{12}d - X\gamma_2}{d^2}, \frac{\partial Y}{\partial u} = \frac{b_{11}d - Y\gamma_1}{d^2}, \frac{\partial Y}{\partial v} = \frac{b_{12}d - Y\gamma_2}{d^2} \tag{9}$$

The two columns of the Jacobian matrix represent the linear

mappings of unit increments along the pixel u and v-axes onto the physical plane. The Euclidean norms of these column vectors therefore define the local linear scales in the two principal directions:

$$s_x(u,v) = \|J(u,v)[1,0]^{\mathrm{T}}\|_2, s_y(u,v) = \|J(u,v)[0,1]^{\mathrm{T}}\|_2 \tag{10}$$

In the subsequent crack width calculation, the local width direction $\mathbf{t} = (\cos\phi, \sin\phi)^{\mathrm{T}}$ is estimated along the crack centerline at each parametric position (u,v). Edge points are then located on both sides of the centerline to obtain the corresponding pixel spacing $d_{\mathrm{pix}}(u,v)$. Using the previously defined local metric, this pixel spacing is converted to physical length through the normal-direction millimeters-per-pixel factor:

$$s_t(u,v) = \|J(u,v)\mathbf{t}\|_2 \tag{11}$$

yielding the physical crack width

$$w(u,v) = s_t(u,v)d_{\mathrm{pix}}(u,v) \tag{12}$$

Since (X,Y) are already expressed in absolute metric units, no additional calibration constant is required. The scale thus possesses an inherent physical dimension, providing a unified and differentiable geometric channel for high-precision quantification of crack width.

4. CrackMamba-net for crack segmentation

To address the dual requirements of fine crack segmentation and subsequent geometric quantification in UAV-based bridge inspection scenarios, a dedicated semantic segmentation network, termed CrackMamba-Net, is developed for the accurate delineation of subtle cracks. The network is designed with the primary objective of preserving the structural coherence of crack masks while enhancing adaptability to scale variations. On the premise of accurate boundary localization, particular emphasis is placed on suppressing interference from complex background textures, shadow effects, and structural edges. Unlike general-purpose segmentation models that mainly focus on pixel-level classification accuracy, CrackMamba-Net prioritizes the spatial continuity and morphological consistency of segmentation results, thereby reducing the risk of error propagation in subsequent geometric operations such as skeleton extraction and normal-direction width measurement. To this end, an encoder–decoder paradigm is adopted as the backbone architecture, and visual state space modeling, multi-scale contextual aggregation, and attention-guided feature fusion are incorporated at key stages to balance global dependency modeling with local detail fidelity. The overall architecture of CrackMamba-Net is first described, followed by a detailed introduction of its core modules and implementation details.

4.1. Overall architecture

Accurate segmentation of fine cracks is critical for the health assessment and maintenance of concrete structures. However, cracks are typically narrow, irregular in shape, and randomly distributed, often embedded in complex and variable background textures. Conventional convolutional networks face clear limitations in capturing edge details, identifying crack regions, and distinguishing noise, which can lead to missed detections and false positives. In practical engineering scenarios, significant scale variations further complicate the task, as single-scale feature extraction cannot effectively handle cracks of varying widths and orientations. To address these challenges efficiently, this study proposes a lightweight segmentation network tailored for slender and elongated cracks. The network follows the encoder-decoder architecture of U-Net, as shown in Fig. 8. In quantitative measurement tasks, the topological continuity of the crack mask is critical: local discontinuities in the mask can lead to false endpoints during the skeletonization process, severely disrupting subsequent principal direction calculation based on PCA. To address this, the network encoder utilizes a global-local visual state space module (GL-VSS) as the core unit, which models local edge textures and long-range dependencies through a stepwise interaction mechanism. This approach enhances global consistency while maintaining sensitivity to fine details. Convolutional Block Attention Modules (CBAM) [45] are embedded in each skip connection to adaptively recalibrate both channel and spatial

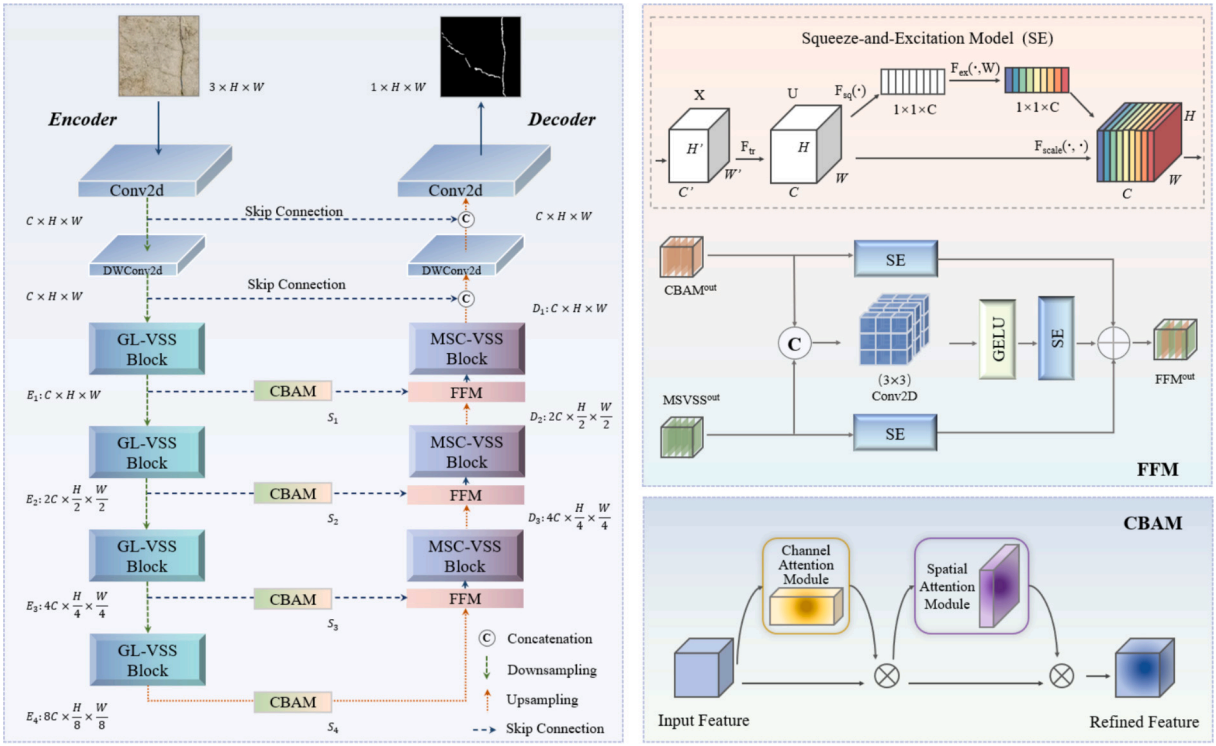


Fig. 8. Architecture of the proposed CrackMamba-Net for crack segmentation.

dimensions, emphasizing crack-relevant responses and suppressing background interference. On the decoder side, multi-scale convolutional visual state space modules (MSC-VSS) are introduced at each resolution level. A feature fusion module (FFM) is then used to integrate shallow high-resolution details from skip connections with the current decoding features. This design enables progressive resolution recovery while preserving semantic continuity and sharp boundary delineation.

In this study, let H and W denote the height and width of the input image (default H=W=512), and let C represent the base number of channels (default $C = 32$). The network input is an RGB image $X \in \mathbb{R}^{3\times H\times W}$. In the input stem, a lightweight two-layer structure is employed. First, a 1×1 convolution performs channel projection, mapping $3\times H\times W$ to $C\times H\times W$. Then, a depthwise separable 3×3convolution is applied without changing the spatial resolution to enhance edge and texture representation, yielding the feature map $X^{'} \in \mathbb{R}^{C\times H\times W}$, which serves as the input to the first GL-VSS module. The encoder follows a four-level hierarchical structure. The first level standardizes the channel dimension to C while keeping the resolution unchanged. The following three levels perform progressive downsampling, where the resolution is halved and the channel number doubled at each stage. Accordingly, the feature dimensions at each level are: $E_1 : C\times H\times W$,$E_2 : 2C\times \frac{H}{2}\times \frac{W}{2}$, $E_3 : 4C\times \frac{H}{4}\times \frac{W}{4}$, $E_4 : 8C\times \frac{H}{8}\times \frac{W}{8}$. Each level's feature is recalibrated using the CBAM to obtain skip features $S_i(i = 1 \sim 4)$, which share the same dimensions as their corresponding E_i. The decoder begins from the lowest-resolution feature $S_4(8C\times \frac{H}{8}\times \frac{W}{8})$, where an initial refinement is performed to generate the first decoding feature at the same scale. The resolution is then progressively restored in an upward manner. Specifically, at the first upsampling stage, the low-scale feature is upsampled to one-fourth resolution ($4C\times \frac{H}{4}\times \frac{W}{4}$)and fused with the corresponding skip feature S_3 in the FFM. The channel dimension is standardized to $4C$, followed by refinement using the MSC-VSS module to obtain decoding features at $\frac{H}{4}\times \frac{W}{4}$. At the second upsampling stage, the feature is raised to half resolution ($\frac{H}{2}\times \frac{W}{2}$), fused with S_2, normalized to $2C$ channels, and further refined by MSC-VSS, generating the decoding feature at that scale. At the third upsampling stage, the feature is restored to full resolution ($H\times W$), fused with $S_1(C\times H\times W)$, normalized to C , and processed again by MSC-VSS to enhance semantic consistency and boundary precision. To further strengthen boundary and fine-texture representation at full resolution, two shallow skip connections are introduced between the stem and the prediction head. The two intermediate convolutional outputs from the stem are concatenated with the shallowest decoding feature at full resolution. After lightweight alignment and normalization to channels, the feature is refined by a depthwise 3×3 convolution (DWConv). Finally, a 1×1 linear projection followed by a Sigmoid activation produces the binary crack mask with a size of $1 \times H \times W$.

4.2. Two-dimensional selective scan module

SS2D (2D Selective Scan) is a key component of the GL-VSS Block and MSC-VSS Block proposed in this study. Originally introduced in the Visual State Space Model (VMamba) [46], SS2D extends the one-dimensional Mamba state space model to two-dimensional visual scenarios. As illustrated in Fig. 9, the core idea of the SS2D module is to sequentially scan the feature map along four diagonal directions, converting the two-dimensional feature map into four one-dimensional sequences. Each sequence is processed by an independent S6 unit (VMamba's selective scan state space block) to capture long-range dependencies; the outputs are then fused and reconstructed into a single 2D feature map. This multi-directional scanning, sequential modeling, and two-dimensional reconstruction process efficiently captures global contextual information while preserving spatial structure. It complements local detail representation and thereby enhances the network's ability to characterize fine cracks and improve segmentation accuracy.

4.3. Encoder architecture with global–local VSS block

To simultaneously enhance fine local textures and global spatial dependencies during the encoder stage, this study proposes the global-local visual state space module (GL-VSS), as shown in Fig. 10(a). The module follows the hierarchical processing concept of the Transformer [47]. It first applies feature normalization and linear projection to stabilize training and integrate channel representations. Then, the 2D selective scan (SS2D) replaces self-attention to model long-range dependencies with near-linear computational complexity. A parallel convolutional branch is introduced to maintain high sensitivity to edges and textures. Finally, a multi-layer perceptron (MLP) performs nonlinear

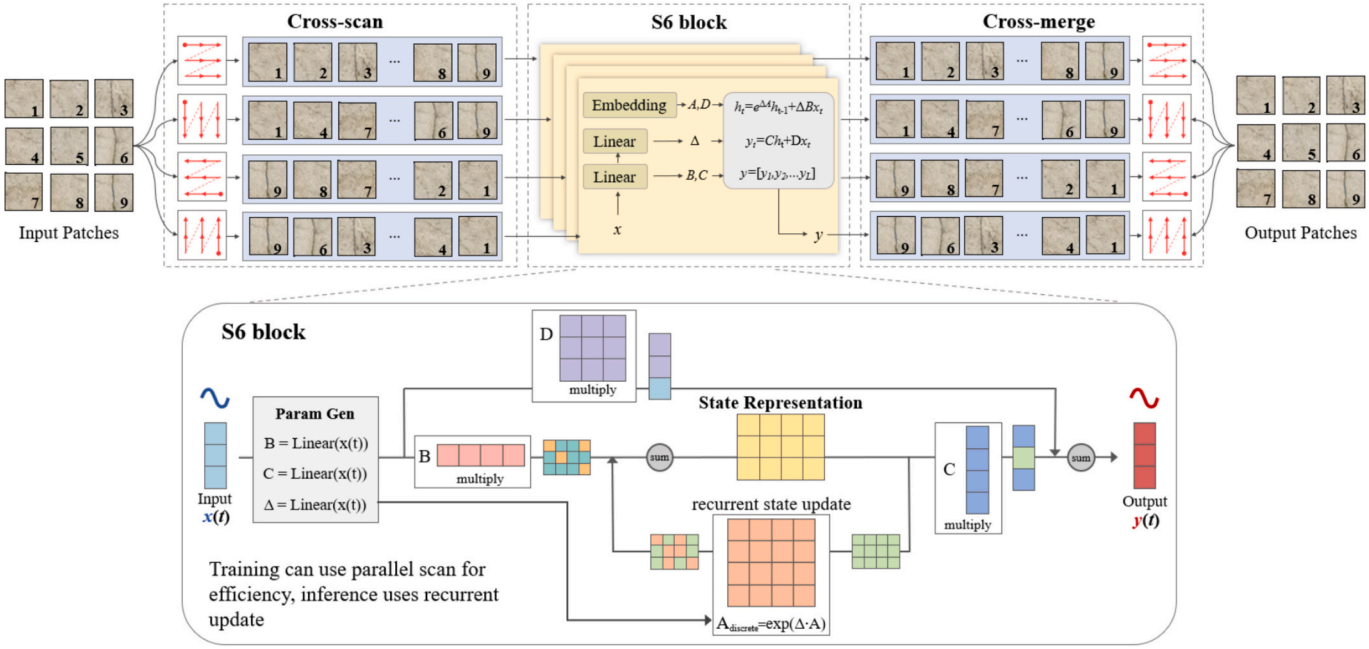


Fig. 9. Illustration of 2D Selective Scan (SS2D).

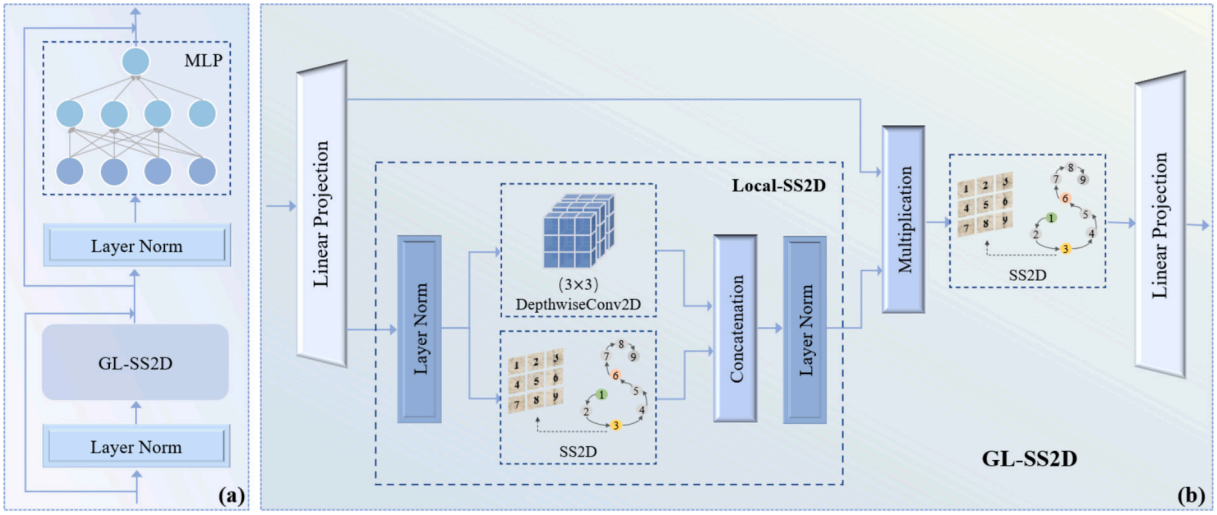


Fig. 10. Illustration of (a)GL-VSS Block, (b)GL-SS2D module.

integration and channel reconstruction to further enhance the representation capability for complex features. First, the input feature X is linearly projected into a higher-dimensional space and divided into two sub-feature paths:

$$X_1, X_2 = \mathit{LinearProjection}(X) \tag{13}$$

where X_1 is sent to the Local-SS2D submodule for local feature extraction, and X_2 is directly processed by the SS2D module to capture global contextual information. In the Local-SS2D submodule, the feature X_1 is first normalized by LayerNorm and then evenly split along the channel dimension. Half of the channels undergo a 3×3 depthwise convolution (DepthwiseConv2D) to exploit the strong capability of convolution for learning local spatial textures, while the other half are processed by the SS2D operation to incorporate the efficient global modeling advantages of the Mamba state space model. This process can be expressed as follows:

$$X_{1a}, X_{1b} = \mathrm{SplitChannels}(\mathrm{LayerNorm}(X_1)) \tag{14}$$

$$X_{conv} = \mathrm{DepthwiseConv2D}(X_{1a}) \tag{15}$$

$$X_{ss2d} = \mathrm{SS2D}(X_{1b}) \tag{16}$$

$$X_{\mathrm{local}} = \mathrm{LayerNorm}(\mathrm{Concat}(X_{conv}, X_{ss2d})) \tag{17}$$

The proposed design allows convolution operations to precisely extract local crack details while the SS2D module models long-range dependencies across the entire image. The two feature representations are then concatenated along the channel dimension to form a unified feature map, significantly enhancing the network's overall feature learning capability. Next, the locally enhanced feature X_{local} and the global feature path output $X_{\mathrm{global}} = \mathrm{SS2D}(X_2)$ are fused through element-wise multiplication to emphasize key features in the crack regions:

$$X_{\mathrm{fused}} = X_{\mathrm{local}} \odot X_{\mathrm{global}} \tag{18}$$

where $\odot$denotes element-wise multiplication. The resulting fused feature X_{fused} combines local detail with global contextual information, effectively improving sensitivity to subtle crack structures. Subsequently, the fused feature undergoes a second SS2D operation for refined global context modeling, followed by a linear projection layer that outputs the final feature representation:

$$Y = \mathrm{LinearProjection}\big(\mathrm{SS2D}\big(X_{\mathrm{fused}}\big)\big) \tag{19}$$

This second SS2D step further enriches high-level global dependencies, while the linear projection restores the feature to an appropriate channel dimension for subsequent nonlinear transformation in the MLP layer. The MLP then applies two linear mappings with activation to produce a feature output of higher expressive capacity.

Through this hierarchical design, the GL-VSS Block effectively balances local detail extraction and global spatial dependency modeling. Its multi-stage interactions reduce redundant information and substantially improve segmentation performance in concrete crack detection tasks.

4.4. Decoder architecture with multi-scale convolutional VSS block

To preserve global contextual awareness during decoding and achieve fine-grained representation of cracks at multiple scales, this study proposes the multi-scale convolutional visual state space module (MSC-VSS), as illustrated in Fig. 11. The core idea is to first stabilize and reorganize the input features, then establish two complementary processing paths. The 2D selective scan (SS2D) captures long-range dependencies with near-linear computational cost, while a depthwise separable convolution enhances local texture and boundary responses. On this basis, parallel multi-scale convolutional branches are introduced to cover diverse receptive fields. Weighted fusion of these branches enables adaptive representation of cracks with varying widths and orientations. Finally, nonlinear integration and channel restoration are performed to produce refined decoding features. This design enhances detail fidelity and scale robustness while maintaining global consistency.

First, the input feature F_i^d undergoes Layer Normalization and Linear Projection, yielding the intermediate feature representations:

$$U_i = \mathrm{DepthwiseConv2D}\big(\mathrm{LayerNorm}\big(F_i^d\big)\big) \tag{20}$$

$$V_i = \mathrm{SS2D}\big(\mathrm{LayerNorm}\big(F_i^d\big)\big) \tag{21}$$

where SS2D captures global dependencies in near-linear time, ensuring the visibility of weak-texture cracks that span across textured or shadowed regions. In parallel, DepthwiseConv2D processes the features along the channel dimension to balance feature distribution and perform pre-filtering, thereby reducing the influence of noise on local crack features. Both outputs are activated by SiLU and fused through element-wise addition, producing the intermediate representation:

$$\widehat{F}_i^d = U_i + \sigma(V_i) \tag{22}$$

where $\sigma(\cdot)$ denotes the activation function. The resulting feature $\widehat{F}_i^d$

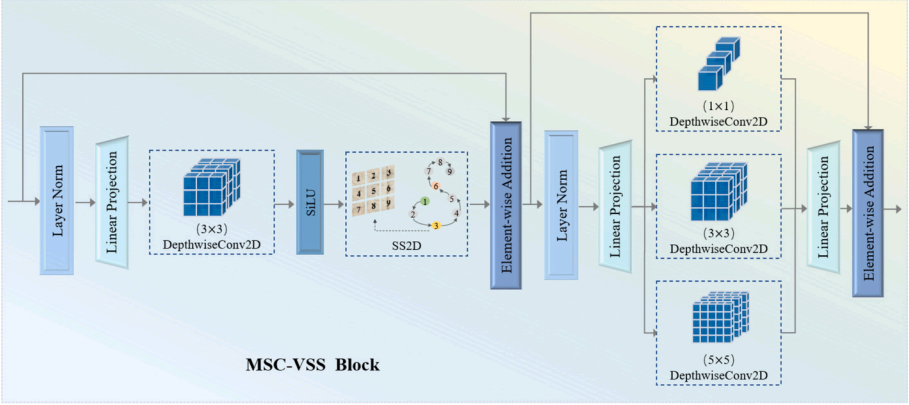


Fig. 11. Illustration of MSC-VSS Block.

combines local texture balancing with preliminary global dependency information. Subsequently, the MSC-VSS Block applies another Layer Normalization and Linear Projection to $\widehat{F}_i^d$, generating the tensor M_i used for parallel multi-scale convolution operations:

$$M_i = \text{LinearProjection}\left(\text{LayerNorm}\left(\widehat{F}_i^d\right)\right) \tag{23}$$

To enhance the network's robustness to variations in crack width, the MSC-VSS incorporates a three-branch parallel depthwise separable convolution structure within its multi-scale feedforward module. Convolution kernels of sizes 1 × 1, 3 × 3, and 5 × 5 are used to capture features at different spatial scales. The 1 × 1 and 3 × 3 kernels are effective in detecting subtle grayscale variations of narrow cracks, thereby improving the network's sensitivity to fine-width cracks. The 5 × 5 kernel, on the other hand, covers a larger contextual area, making it suitable for extracting features of wider cracks and their surrounding damaged regions. The outputs of the three convolutional branches are fused through element-wise addition and then projected back to the original channel dimension using a linear projection to obtain the final output:

$$F_{i+1}^d = \text{LinearProjection}\left(M_i^{(1)} + M_i^{(3)} + M_i^{(5)}\right) \tag{24}$$

This parallel multi-scale convolution structure not only extracts and integrates effective crack edge information across multiple scales but also suppresses noise responses caused by pseudo-cracks such as surface pits or impurities, thereby improving segmentation accuracy. In summary, the MSC-VSS Block achieves efficient multi-level crack feature fusion by combining channel-wise feature balancing, global dependency scanning, and multi-scale convolution in parallel. It preserves global semantic information while maintaining sensitivity to crack details and scale variations, ensuring that the decoder retains high accuracy and robustness during progressive spatial resolution recovery.

4.5. Attention-driven skip-connection fusion via CBAM and FFM

To compensate for detail loss caused by upsampling during decoding and to better constrain crack boundaries, a channel-spatial attention and feature fusion mechanism is introduced at each skip connection, as shown in Fig. 8. Specifically, the Convolutional Block Attention Module (CBAM) [45] first performs two-stage recalibration on the high-resolution features from the encoder. The channel attention component adaptively emphasizes crack-related channels through global pooling and a shared multi-layer perceptron (MLP), while the spatial attention component generates weight maps using average pooling and convolution to highlight target regions and suppress background noise, producing the enhanced skip feature F_{CBAM}. Subsequently, the feature fusion module (FFM) efficiently merges F_{CBAM} with the current decoding feature $F_{\text{MSC-VSS}}$. Both feature streams are first processed by 3 × 3 depthwise separable convolutions and GELU activation to strengthen representation. The fused features are then adaptively reweighted through the Squeeze-and-Excitation (SE) mechanism [48], followed by element-wise addition and a 1 × 1 linear projection to restore the channel dimension, yielding the final fused output F_{FFM}. Through the combined effect of CBAM and FFM, the network effectively leverages multi-level feature information, achieving more robust and accurate segmentation predictions under complex backgrounds and varying crack morphologies.

4.6. Loss function

Crack segmentation is formulated as a pixel-level binary classification task, where each pixel is classified as either crack or background. Since cracks occupy a relatively small portion of the image, the dataset is highly imbalanced between positive and negative samples. To address this, the network employs a combination of binary cross-entropy loss (BCELoss) [49] and Dice loss (DiceLoss) [50]. The BCELoss accurately measures classification errors between positive and negative samples, while the DiceLoss optimizes the overlap between predictions and ground truth, effectively reducing bias caused by class imbalance. The two losses complement each other, enabling the model to distinguish fine cracks accurately while maintaining robust overall predictions. The final loss function is defined as

$$L_{\text{BceDice}} = 0.5L_{\text{BCE}} + 0.5L_{\text{Dice}} \tag{25}$$

where L_{BCE} denotes the binary cross-entropy loss and L_{Dice} denotes the Dice coefficient loss, given by

$$L_{\text{BCE}} = -\frac{1}{N}\sum_{i=1}^{N}\left[y_i \log(p_i) + (1-y_i)\log(1-p_i)\right] \tag{26}$$

$$L_{\text{Dice}} = 1 - \frac{2\sum_{i=1}^{N} y_i p_i}{\sum_{i=1}^{N} y_i + \sum_{i=1}^{N} p_i} \tag{27}$$

where y_i represents the ground truth label, p_i is the predicted value, and N is the total number of samples.

4.7. Experimental results and analysis

To systematically validate the crack segmentation performance and practical applicability of the proposed CrackMamba-Net in complex engineering scenarios, this section conducts a comprehensive experimental study encompassing dataset construction, experimental settings, evaluation metrics, and comparative analyses. The experimental design not only examines pixel-level segmentation accuracy but also

emphasizes the stability and robustness of the model under weak contrast, fine-scale cracks, and complex background conditions, thereby assessing the extent to which the segmentation results support subsequent geometric quantification tasks. Quantitative and qualitative comparisons are performed against multiple mainstream segmentation networks under a unified dataset and evaluation protocol, and ablation studies are further conducted to analyze the contributions of the core modules. This section aims to provide an integrated assessment of CrackMamba-Net in terms of accuracy, structural coherence, and computational efficiency. The datasets and their construction are first introduced, followed by the experimental configuration, evaluation metrics, and result analysis.

4.7.1. Dataset

This study employs various data collection methods, including UAVs, cameras, and smartphones, to capture crack field data on key components of the Jiyin Bridge, such as the soffit, flange plates, and piers, under complex background conditions. To ensure the high quality and reliability of the dataset, strict annotation protocols were established. All images were pixel-level annotated by three researchers with civil engineering backgrounds using the open-source tool LabelMe. Given the narrow and irregular nature of the cracks, a fine polygonal annotation strategy was employed to closely trace the crack edges, instead of using rough rectangular bounding boxes. The annotation results were then converted into binary masks. For quality control, a cross-validation mechanism was introduced, with approximately 10% of the data randomly selected for re-annotation by a different annotator. The inter-annotator agreement, quantified by the average mean intersection over union (mIoU), reached 0.85, indicating high consistency. All samples were finally reviewed by a senior structural engineer to establish the Ground Truth. Based on this process, a dataset containing 7548 images was created, named "Jiyincrack" (Table 1). For effective model training and evaluation, the "JiyinCrack" dataset was divided into training and testing subsets using an 8:2 ratio, where the training set contains 6038 images and the testing set includes 1510 images. All images share the same resolution of 512 × 512 pixels. Although the dataset was collected from a single location at the Jiyin Bridge, the sample construction focused on ensuring Intra-source Diversity. The collection areas include different structural parts of the bridge, such as the pier webs, box girder soffits, and lateral flange plates, and encompass various complex backgrounds, including concrete surface whitening, water stains, moss coverage, and strong light/shadow interference. This diverse sample distribution ensures that CrackMamba-Net learns robust crack features and avoids overfitting to a single texture.

4.7.2. Experimental setup

This study employs the Adam optimizer for network training. Specifically, three key hyperparameters are initialized as follows: $\beta = (0.9, 0.999)$, weight decay $= 1\times10^{-4}$ and learning rate $= 1\times 10^{-4}$. The model is trained for 50 epochs, and to prevent out-of-memory issues, the batch size is set to 2. The training process is conducted on a Windows 10 operating system using the PyTorch framework. The software environment is configured as follows: Python 3.8.0, PyTorch 1.8.1, CUDA 11.1, and CUDNN 8.0.4. The hardware platform used in this study is an NVIDIA GeForce RTX 3060 GPU with 12 GB of memory.

4.7.3. Evaluation metrics

To evaluate the performance differences between CrackMamba-Net and other networks, several commonly used quantitative metrics were adopted, including precision, recall, F1-score, and mean Intersection over Union (mIoU). The calculation formulas are as follows:

$$Precision = \frac{TP}{TP + FP} \tag{28}$$

$$Recall = \frac{TP}{TP + FN} \tag{29}$$

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall} \tag{30}$$

$$mIoU = \frac{1}{N}\sum_{i=1}^{N} \frac{TP}{TP + FP + FN} \tag{31}$$

Precision measures the classification accuracy of positive samples, representing the proportion of correctly predicted crack pixels among all pixels identified as cracks. Recall assesses the model's ability to capture positive samples, reflecting the proportion of actual crack pixels that are successfully detected. Since precision and recall emphasize different aspects of performance and are both influenced by the classification threshold, the F1-score provides a balanced measure by computing their harmonic mean. The mean Intersection over Union (mIoU) is a standard metric in semantic segmentation that quantifies the overall overlap between predictions and ground truth. For the binary task of crack segmentation, only two classes are involved: crack and background. The use of these metrics enables a comprehensive assessment of model performance in both crack detection and classification. In the equations above, TP, FP, and FN denote true positives, false positives, and false negatives, respectively.

4.7.4. Visual result analysis

In this study, four representative samples from the self-developed JiyinCrack dataset were selected, each differing in texture, brightness, and crack morphology. Seven widely used segmentation networks, including Fast-SCNN [51], U-Net [52], PSPNet [53], DeepLabV3+ [54], SegNet [55], VM-UNet [56], and SegFormer-B5 [57], were compared with the proposed CrackMamba-Net, and their visual results are shown in Fig. 12. Overall, CrackMamba-Net maintains superior global connectivity and detail fidelity under weak contrast, fine cracks, and complex backgrounds. In Samples 1 and 3, which feature low contrast and rough textures, it yields clear mask boundaries and continuous centerlines with minimal breakage or geometric drift. By contrast, U-Net, PSPNet, and SegNet show blurred edges, local discontinuities, and a tendency to overestimate widths in narrow regions. For the slender and tortuous crack in Sample 2, CrackMamba-Net preserves near-single-pixel continuity at high-curvature turns, whereas DeepLabV3+ and VM-UNet often produce gaps or holes in noisy areas, and SegFormer-B5 occasionally misclassifies shadows or texture lines as cracks, indicating sensitivity to local texture interference. In Sample 4, strong background edges and specular reflections cause excessive responses for most methods, leading to false positives and boundary merging. Benefiting from long-range dependency modeling and noise suppression within the state space model, CrackMamba-Net suppresses background structures and preserves thin lines consistent with crack topology. Cross-sample comparisons show smaller fluctuations in line width and better width consistency for the proposed method, which supports accurate skeleton-based normal-direction width measurement. Several comparison methods exhibit over-segmentation or under-segmentation where texture varies strongly, causing boundary swelling or contraction and introducing systematic bias in width estimation.

To visually demonstrate performance differences among the models under extreme conditions, local zoom-ins of particularly challenging regions are provided in Fig. 12, as highlighted by the red boxes. As shown, in the fine crack branches of samples 1 and 2, the comparison models (such as SegFormer-B5 and U-Net) exhibit noticeable pixel-level

Table 1
Details of the experiment datasets.

Dataset	Image size for training	Number of images for training	Image size for test	Number of images for test
JiyinCrack	512 × 512	6038	512 × 512	1510

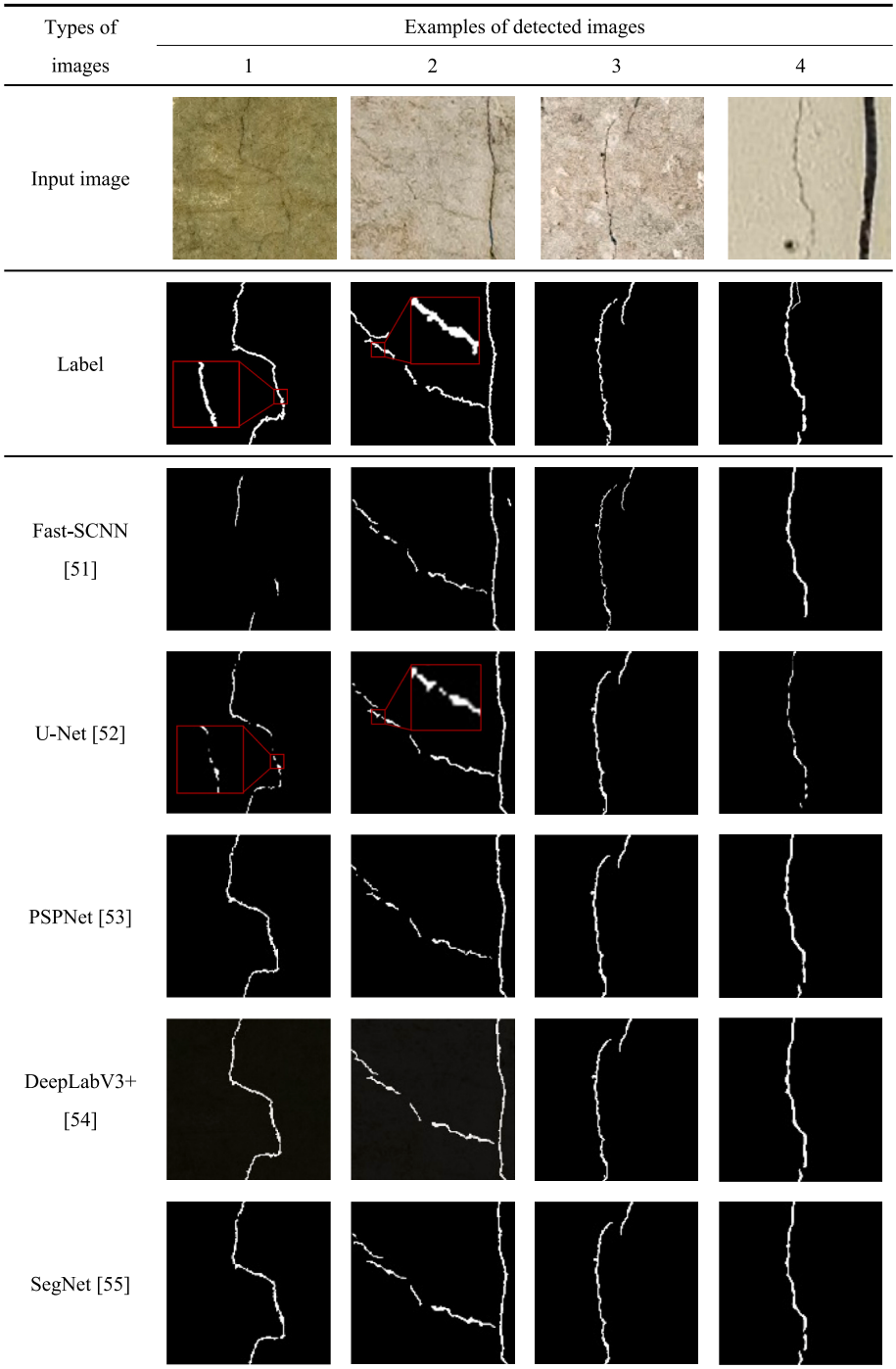


Fig. 12. Visualized results of different networks.

breakage and structural discontinuities, which can lead to erroneous skeleton extraction in subsequent geometric quantification. In contrast, CrackMamba-Net, with its unique long-range context modeling capability, successfully "bridges" these weak texture regions, maintaining the topological continuity of the crack skeleton and accurately restoring fine edge details. This continuity advantage is crucial for ensuring the accuracy of normal-based crack width measurement. However, it is noteworthy that under extreme conditions, such as long-distance imaging or high-texture backgrounds, CrackMamba-Net still faces challenges in maintaining the mask integrity for very fine, "hairline" cracks. This limitation somewhat restricts the quantification accuracy in such scenarios.

In summary, CrackMamba-Net balances global continuity and local detail, shows stronger resistance to noise and background interference, and provides stable, verifiable crack masks for subsequent geometric quantification.

4.7.5. Comprehensive analysis of results

Table 2 presents the quantitative performance of eight networks evaluated under the same dataset and testing protocol. Overall, CrackMamba-Net achieves the best performance across all key accuracy metrics. Its mIoU reaches 88.79%, indicating superior pixel-level distinction between cracks and background with the highest regional overlap. The precision of 85.23% demonstrates effective suppression of

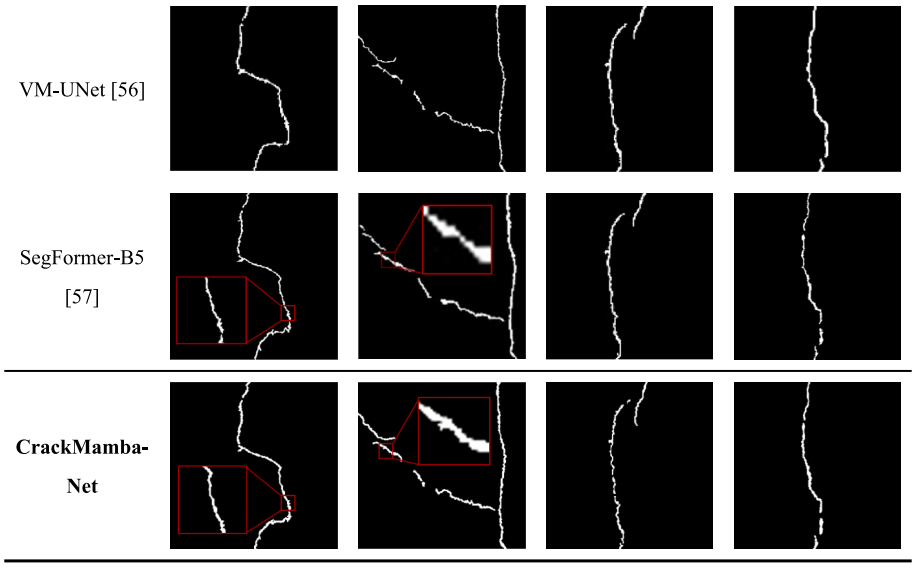


Fig. 12. (*continued*).

Table 2
Performance comparison of different methods.

Network	mIoU (%)	Precision (%)	Recall (%)	F1-score (%)	Params (M)	FLOPs (G)
Fast-SCNN	69.97	63.84	65.49	64.14	**1.14**	**9.17**
U-Net	73.76	74.97	71.43	72.15	7.77	51.32
PSPNet	75.86	75.35	72.87	73.41	42.57	75.97
DeepLabV3+	78.98	78.11	71.62	74.86	59.34	81.63
SegNet	80.64	80.26	73.25	75.59	29.44	147.82
VM-UNet	81.97	81.04	74.68	77.37	27.43	54.11
SegFormer-B5	84.41	82.68	79.65	81.65	82.71	112.73
CrackMamba-Net	**88.79**	**85.23**	**82.63**	**83.92**	47.82	67.51

Notes. (1) Params denotes the number of model parameters (millions, M); FLOPs denotes computational complexity (billions of floating-point operations, G).

false positives, as non-crack structures such as texture lines, shadows, and joints near boundaries are less frequently misclassified. The Recall of 82.63% reflects more complete coverage of actual crack pixels with fewer missed detections in narrow or low-contrast regions. The F1-score of 83.92% confirms a balanced overall performance between detection accuracy and completeness. It is noteworthy that VM-UNet, one of the baseline models, represents the Mamba family of state space model architectures, while U-Net, PSPNet, DeepLabV3+, SegNet, and Fast-SCNN belong to traditional CNN-based methods, and SegFormer-B5 represents the Transformer-based paradigm. Even when compared with VM-UNet, which shares the same Mamba backbone and long-range dependency modeling capability, CrackMamba-Net achieves higher mIoU, Precision, Recall, and F1-score. This demonstrates that the proposed multi-scale contextual aggregation, boundary-sensitive constraints, and lightweight noise suppression strategies enhance both global coherence and local boundary accuracy, reducing missed detections of fine cracks while suppressing texture-induced false detections. In terms of computational efficiency, CrackMamba-Net contains 47.82 M parameters and 67.51G FLOPs, placing it within a moderately complex yet deployable range. Compared with larger networks, it achieves higher segmentation accuracy with fewer parameters and lower computational cost, demonstrating an effective balance between precision and efficiency. While its computational load is slightly higher than that of ultra-lightweight models, it delivers significantly better regional consistency and boundary completeness, making it particularly suitable for providing reliable crack masks in subsequent geometric quantification.

In practical engineering applications, UAV-based inspection often involves the collection of large volumes of high-resolution images, and the processing efficiency of subsequent algorithms directly impacts the inspection cycle time and response speed. To evaluate the suitability of CrackMamba-Net for offline processing scenarios, its inference speed and resource usage were assessed in a standard mobile workstation environment equipped with an NVIDIA GeForce RTX 3060 GPU. The results indicate that, thanks to the Mamba architecture's advantage in linear computational complexity, CrackMamba-Net achieves an inference speed of 52 FPS with an input size of 512 × 512, and a memory usage of 1350 MB. This means processing 1000 inspection images takes approximately 20 s. While real-time inference on high-resolution images demands significant onboard computational resources, there is still room for improvement in the frame rate on low-power embedded platforms.

Overall, CrackMamba-Net achieves a balance between controlled inference costs and exceptional feature representation capability. It consistently produces segmentation results with high connectivity, low false detection rates, and low miss rates, providing a solid and reliable visual foundation for subsequent millimeter-level crack geometric quantification.

4.8. Ablation study and module effectiveness analysis

To systematically validate the effectiveness of each core component in CrackMamba-Net and quantify the specific contributions of the Global-Local Visual State Space Module (GL-VSS), Multi-Scale Convolutional Visual State Space Module (MSC-VSS), and the attention fusion mechanisms (CBAM & FFM) to crack segmentation performance, a step-by-step ablation study was conducted on the JiyinCrack dataset.

4.8.1. Experimental setup

In this experiment, a U-shaped architecture built upon the basic Visual State Space (VSS) unit is adopted as the baseline. Using a controlled-variable design, the proposed modules are introduced sequentially, yielding four model variants for quantitative evaluation:

Model A (Baseline): Consists of an encoder and decoder built solely with the basic VSS module, without the multi-scale branches and attention enhancement mechanisms tailored for crack features.

Model B (+ GL-VSS): Introduces the Global-Local Visual State Space Module (GL-VSS) at the encoder stage, replacing the basic VSS block. This aims to verify the effectiveness of collaborative modeling between

local convolution and global scanning.

Model C (+ MSC-VSS): Based on Model B, MSC-VSS is introduced at the decoder stage to test the adaptability of multi-receptive field feature aggregation to variations in crack scale.

Model D (CrackMamba-Net): The full model, which integrates CBAM attention mechanism and FFM feature fusion module at the skip connections, in order to evaluate their contributions to feature alignment and suppression of complex background noise.

4.8.2. Results and analysis

The quantitative results of the ablation study are summarized in Table 3. The data indicate a steady improvement in the network's performance metrics with the sequential introduction of each innovative module, validating the rationality of the architecture design and the complementarity between the modules.

Effectiveness of Global-Local Collaborative Modeling (GL-VSS): Compared to the baseline model, Model B shows a significant performance boost after incorporating the GL-VSS module. This improvement is primarily due to the dual-branch design of the GL-VSS module: one branch captures long-range dependencies via the SS2D scanning mechanism with near-linear complexity, maintaining global continuity of the crack skeleton; the other branch strengthens the response to local edge textures through parallel convolution. The combination of these two mechanisms effectively resolves the issue of detail loss in fine crack edges when using a pure SSM model, significantly improving the topological integrity of the segmentation results.

Contribution of Multi-Scale Feature Aggregation (MSC-VSS): The introduction of the MSC-VSS module in the decoder stage (Model C) further enhances the mIoU metric. This module integrates parallel multi-scale depthwise separable convolutions with kernel sizes of 1×1, 3×3, and 5×5, enabling adaptive capture of crack features with varying widths. Experimental results demonstrate that the fusion of multi-scale receptive fields allows the network to maintain more robust segmentation performance when dealing with cracks of varying widths or fine, hairline cracks, effectively reducing breakage or under-segmentation due to scale mismatch.

Noise Suppression in Complex Backgrounds (via CBAM & FFM): The full CrackMamba-Net model (Model D), after integrating CBAM and FFM, achieved optimal performance, with mIoU reaching 88.79% and F1-score reaching 83.92%. The CBAM module applies dual attention recalibration in both channel and spatial dimensions, enabling the model to focus on significant crack regions and automatically suppress background noise such as water stains and oil marks. Meanwhile, the FFM module performs adaptive channel weighting to efficiently fuse high-resolution features from the encoder with semantic features from the decoder. This mechanism ensures that the final output mask maintains clear boundaries while achieving a very low false detection rate (Precision reaching 85.23%).

In summary, the ablation study thoroughly demonstrates that GL-VSS, MSC-VSS, and the attention fusion mechanisms each make unique and indispensable contributions to the superior performance of CrackMamba-Net. The modules complement each other in feature extraction, scale adaptation, and noise suppression, collectively supporting high-precision segmentation of concrete cracks in complex environments.

5. Crack width measurement via skeleton refinement and linear correction

To achieve a stable, repeatable, and engineering-interpretable quantification of concrete crack width, this study establishes a rigorously structured geometric measurement framework based on the binary crack mask obtained from segmentation. Specifically, the crack skeleton is first refined to ensure topological reliability, followed by extraction of the dominant local orientation using principal component analysis (PCA) [58]. A fine-grained weighted least squares (WLS) linear correction is then performed around this orientation, and the intersections between the corrected normal direction and the crack boundaries are determined. The Euclidean distance between the two intersection points defines the pixel-domain crack width. The core motivation of this method is to ensure directional accuracy before performing width computation. Through hierarchical estimation and constraint design, it systematically reduces measurement bias caused by noise, jagged edges, and local curvature, ensuring that the estimated width aligns precisely with the engineering definition of the transverse distance perpendicular to the crack direction.

Skeleton purification is first conducted to prevent segmentation errors from propagating into subsequent geometric computations. After generating the initial skeleton using the classical Zhang-Suen thinning algorithm [59], it is not directly used as the measurement backbone. Instead, two iterative operations, namely short-branch removal and redundant-point deletion, are applied. The former removes isolated branches, starting from the endpoints, with lengths smaller than T_{prune} pixels (set as $T_{prune} = 15$ in this study), which are formed by noise or threshold disturbances, thereby eliminating false orientations. The latter suppresses structural bifurcations caused by excessive local connectivity by counting the number of connections in the 8-neighborhood of each skeleton pixel and constraining it to a maximum of 2. These operations preserve the main connectivity of the skeleton while simplifying its topology, reducing directional ambiguity at the source, and providing a more reliable geometric backbone in both shape and position for robust local statistical analysis. Fig. 13 presents the comparison of crack skeletons before and after refinement.

After obtaining a continuous and redundancy-free skeleton line, the next step is to determine the crack propagation direction to accurately define the orientation of width measurement. For each skeleton point p, principal component analysis (PCA) is performed within its local neighborhood, as illustrated in Fig. 14(a). A $k \times k$ window is extracted from the binary crack mask image, centered at p, with k=7 in this study. This window size is chosen based on the discretization characteristics of the crack skeleton. The value k=7 effectively suppresses quantization noise caused by skeleton pixelation while avoiding the smoothing of geometric features in high-curvature regions due to an excessively large window. The pixel coordinates of all skeleton points within the window are denoted as $\{p_i = (x_i, y_i)\}_{i=1}^{n}$, and their centroid is computed as $\mu = \frac{1}{n}\sum_{i=1}^{n} p_i$. The coordinates are then centralized using $\Delta p_i = p_i - \mu$, forming the data and covariance matrices:

$$X = [\Delta p_1, \Delta p_2, \ldots, \Delta p_n] \in \mathbb{R}^{2\times n}\ ,\ C = \frac{1}{n} XX^{\top} \tag{32}$$

By performing eigen decomposition $Cv_j = \lambda_j v_j (j = 1, 2)$, the unit eigenvector v_1 corresponding to the largest eigenvalue λ_1 represents the dominant local extension direction (local tangent), while the orthogonal

Table 3
Ablation study of component effectiveness on the JiyinCrack dataset.

Methods	GL-VSS	MSC-VSS	CBAM & FFM	Precision(%)	Recall(%)	F1-score(%)	MIoU(%)
Model A (Baseline)	×	×	×	81.48	78.91	79.32	82.28
Model B	✓	×	×	83.64	80.07	81.53	85.32
Model C	✓	✓	×	84.77	81.18	82.79	87.65
Model D (Proposed)	✓	✓	✓	**85.23**	**82.63**	**83.92**	**88.79**

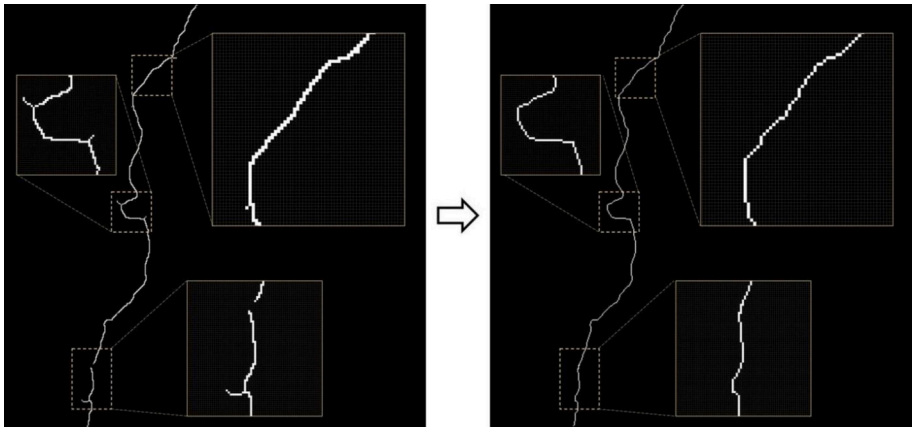


Fig. 13. Comparison of crack skeletons before and after refinement.

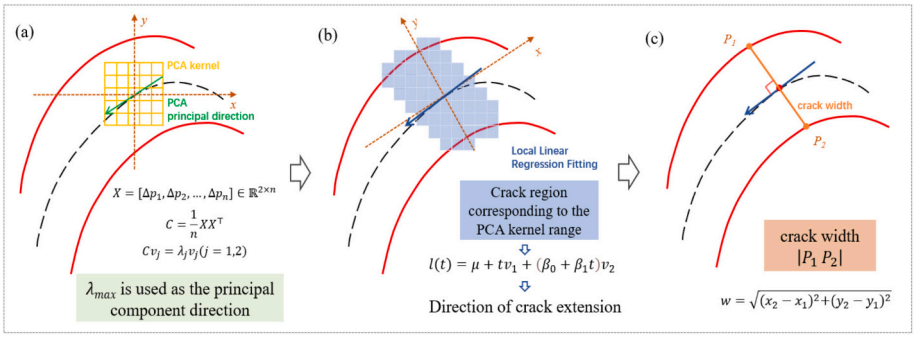


Fig. 14. Workflow of crack width measurement with local linear correction.

unit vector v_2, obtained via a rotation matrix, defines the local normal direction. As a statistical basis for local orientation estimation, PCA identifies the principal geometric trend of the neighborhood according to the major eigenvector, effectively suppressing the influence of isolated outliers and jagged edges, and providing a stable initial orientation for subsequent normal projection. Since relying solely on the PCA-derived principal direction may cause estimation lag in regions with slight curvature or nonuniform texture, a one-dimensional linear refinement is further performed in the local orthogonal coordinate system defined by $\{v_1, v_2\}$. The PCA kernel domain and its corresponding crack edge pixels are projected into one-dimensional coordinates:

$$t_i = v_1^{\mathrm{T}} \Delta p_i, z_i = v_2^{\mathrm{T}} \Delta p_i \tag{33}$$

The slight deflection of the crack extension direction within this local neighborhood can be approximated by a first-order polynomial:

$$z_i = \beta_0 + \beta_1 t_i + \varepsilon_i \tag{34}$$

To balance the emphasis on the central pixel and suppress distant noise, a Gaussian distance weight $w_i = \exp(- \| \Delta p_i \|^2 / \sigma^2)$ is introduced, with the scale parameter set to $\sigma = k/3$ (i.e., $\sigma \approx 2.33$). Based on the properties of the Gaussian distribution, this setting ensures that the weight smoothly decays toward the edges of the window, allowing the direction estimation to effectively leverage the neighborhood context for robust regression while maintaining sensitivity to the local central position. The regression coefficients are then obtained analytically using the weighted least squares criterion:

$$\min_{\beta_0,\beta_1} \sum_{i=1}^{n} w_i (z_i - \beta_0 - \beta_1 t_i)^2, \beta_1 = \frac{\sum_i w_i (t_i - \bar{t})(z_i - \bar{z})}{\sum_i w_i (t_i - \bar{t})^2}, \beta_0 = \bar{z} - \beta_1 \bar{t}, \tag{35}$$

where $\bar{t} = \sum_i w_i t_i / \sum_i w_i, \bar{z} = \sum_i w_i z_i / \sum_i w_i$. The coefficient β_1 describes the local deviation of the crack from the principal direction v_1. Mapping the fitted regression model back to the global coordinate system, the centerline can be expressed as a parametric curve:

$$l(t) = \mu + tv_1 + (\beta_0 + \beta_1 t)v_2, t \in \mathbb{R} \tag{36}$$

The normalized direction vector is then obtained as

$$d_{\mathrm{glob}} = \frac{v_1 + \beta_1 v_2}{\| v_1 + \beta_1 v_2 \|} = \frac{1}{\sqrt{1+\beta_1^2}} v_1 + \frac{\beta_1}{\sqrt{1+\beta_1^2}} v_2 \tag{37}$$

as shown in Fig. 14(b). This vector combines the global variance-maximizing property of PCA with the fine local adjustment captured by least squares regression, providing a precise and robust geometric foundation for crack orientation estimation. Subsequently, by rotating the normalized direction vector $d_{\mathrm{glob}} = (d_x, d_y)^{\mathrm{T}}$ counterclockwise by 90°, an orthogonal unit vector $v_{\perp} = \begin{pmatrix} -d_y \\ d_x \end{pmatrix}$ is obtained, which serves as the reference direction for crack width measurement. At each skeleton point p, intersections with the binary crack mask boundary are identified along both directions of $v_{\perp}$, denoted as $b_1 = (x_1, y_1)$ and $b_2 = (x_2, y_2)$. The crack width w is then defined as the Euclidean distance between these two intersection points, as illustrated in Fig. 14(c):

$$w = \| b_1 - b_2 \|_2 = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} \tag{38}$$

To evaluate the accuracy of the proposed method in determining crack width measurement directions, an angle error test based on manually defined reference orientations [60], was conducted, along with systematic experiments including comparison algorithms. In the dataset established in Section 5, 50 images were randomly selected, and

5 skeleton key points p_i were further randomly chosen from each image as test samples. Three technicians with inspection experience were then invited to independently determine the local crack width direction in the neighborhood of each p_i in high-resolution views with subpixel scaling support. The method involved selecting one boundary pixel A and B on either side of the crack, ensuring the line segment $\overline{AB}$ passes through p_i and is perpendicular to the local texture orientation, thereby defining the manual reference direction $\overrightarrow{AB}$. To eliminate the randomness of individual subjective judgments, the final manual reference truth is defined as the vector average of the unit direction vectors determined by the three experts. The consistency test results show that the average standard deviation of the angles determined by the three experts is less than 3.5°, indicating that the geometrically defined directions have high stability and reliability, providing a solid reference for algorithm evaluation. Based on this, the algorithm proposed in this study was applied to compute the normal direction at the same point p_i and search for the boundary points C and D on both sides, forming the algorithm's estimated direction vector $\overrightarrow{CD}$. The direction estimation error $\Delta\theta$ is defined as the acute angle between the manual reference vector and the algorithm's estimated vector:

$$\Delta\theta = \arccos\left(\frac{\left|\overrightarrow{AB} \bullet \overrightarrow{CD}\right|}{\| \overrightarrow{AB} \| \| \overrightarrow{CD} \|}\right) \tag{39}$$

To quantify the consistency between algorithmic and manual direction judgments, the proposed method was compared with two representative direction estimation strategies at the same sampling points. The shortest distance method [61] defines crack width as the minimum Euclidean distance from a skeleton point to its nearest boundary pixels on both sides, while the orthogonal projection method [62] first estimates the crack orientation and then projects boundaries along its orthogonal direction. All three methods were evaluated on fifty images, each containing five randomly selected main skeleton points (a total of 250 groups). For each point, the direction error $\Delta\theta$ was computed, and the mean, median, range, and standard deviation were statistically analyzed. As shown in Table 4, the proposed method achieved mean and median direction errors of 6.74° and 6.56°, respectively, which are significantly lower than those of the orthogonal projection method (17.77°/15.34°) and the shortest distance method (19.75°/18.73°). The standard deviation of 5.41° also demonstrates better convergence compared with the two baselines (12.55°/12.91°). The error trajectories in Fig. 15 further indicate that the proposed method exhibits smaller amplitude and fewer long-tail deviations, with isolated high values only occurring at points of high curvature or boundary discontinuity. The superior performance arises mainly from two factors: first, PCA provides a low-variance estimation of the principal direction along the skeleton, effectively suppressing local noise and grid effects; second, the distance-weighted least squares refinement performs first-order linear fitting around the dominant direction, allowing it to follow gradual curvature changes without introducing bias. It is worth noting that residual large errors primarily occur in low-contrast or incomplete boundary regions where asymmetry exists on both sides of the normal, which inherently challenges all boundary-based search methods. Overall, the proposed approach achieves lower systematic bias and smaller random fluctuations in direction estimation, providing a more reliable directional prior for subsequent millimeter-scale crack width quantification.

Table 4
Direction error statistics of width measurement direction determination methods.

	Mean /°	Median /°	Max /°	Min /°	Std. dev. /°
Proposed Method	6.74	6.56	38.67	0	5.41
Orthogonal Projection Method	17.77	15.34	56.78	0	12.55
Shortest Method	19.75	18.73	58.69	0	12.91

6. Field validation experiments

To examine the feasibility and metrological reliability of the proposed method in real engineering environments, a series of systematic field validation experiments were conducted, with particular emphasis on the performance of the scale calibration and crack width quantification pipeline under varying observation geometries and complex site conditions. The field experiments were organized around two representative scenarios. First, reference calibration boards with known dimensions were introduced to evaluate the accuracy and stability of four-point laser assisted pixel scale recovery under controlled conditions. Second, UAV-based measurements were performed on an in-service bridge structure to comprehensively verify the overall consistency and transferability of the integrated workflow, including scale injection, crack segmentation, and geometric quantification, in practical engineering settings. The experiments were carried out across multiple viewing distances and imaging poses, and manual contact-based measurements were used as benchmarks for comparison, enabling a thorough assessment of the measurement performance in real-world applications. The following sections first present the field evaluation results of pixel scale estimation based on four-point laser calibration, and then describe the crack detection and width quantification experiments conducted on the actual bridge.

6.1. Experimental evaluation of pixel scale estimation via four-point lasers

To verify the effectiveness and robustness of the proposed method in recovering pixel-to-physical scale under different imaging poses, a field experiment was conducted at the Jiulonghu Campus of Southeast University using a calibrated size reference panel, as shown in Fig. 16. The panel was a flat wall-mounted reference card containing certified strip widths of 0.8, 1.6, 4, 10, and 20 mm. The experimental platform consisted of a DJI M300 RTK equipped with a Zenmuse H20 camera and a four-point laser ranging device. Both the camera and the four ranging channels were synchronized by a common trigger to ensure one-to-one correspondence between each image frame and its four distance observations. To systematically assess performance under different observation geometries, three working distances (2 m, 4 m, and 6 m) and two imaging poses (frontal 0° and oblique 45°) were tested. During data collection, the laser emission directions were finely adjusted through the gimbal and steering mechanism so that the four red laser spots were evenly distributed within the ROI with sufficient spread. The pixel coordinates of the laser spots were extracted within the fixed ROI using zero-mean normalized cross-correlation (ZNCC). Combined with the four measured distances and inter-beam angles, the planar four-point reference geometry was reconstructed. The normalized direct linear transform (DLT) was then applied to estimate the homography matrix, from which the Jacobian matrix was derived to compute the position-dependent local length metric. This metric provided the local pixel-to-physical scale factor (unit: mm·px^{-1}) for each strip position. The strip width was measured by applying Otsu thresholding [63] for segmentation, followed by boundary alignment and pixel-distance measurement along the normal direction. The resulting pixel width was then multiplied by the local length metric to obtain the physical crack width.

Table 5 summarizes the laser ranging and width estimation results obtained under three working distances and two imaging poses to evaluate the stability of the proposed scaling framework against geometric variation. The four-channel laser readings were highly consistent at each distance and pose. For example, at 2 m, 4 m, and 6 m, the range dispersion among the four beams was approximately 1–3 mm, corresponding to a relative deviation below 0.05%, confirming the effectiveness of synchronous triggering and the coplanarity assumption of the

Orthogonal Projection Method	17.77	15.34	56.78	0	12.55

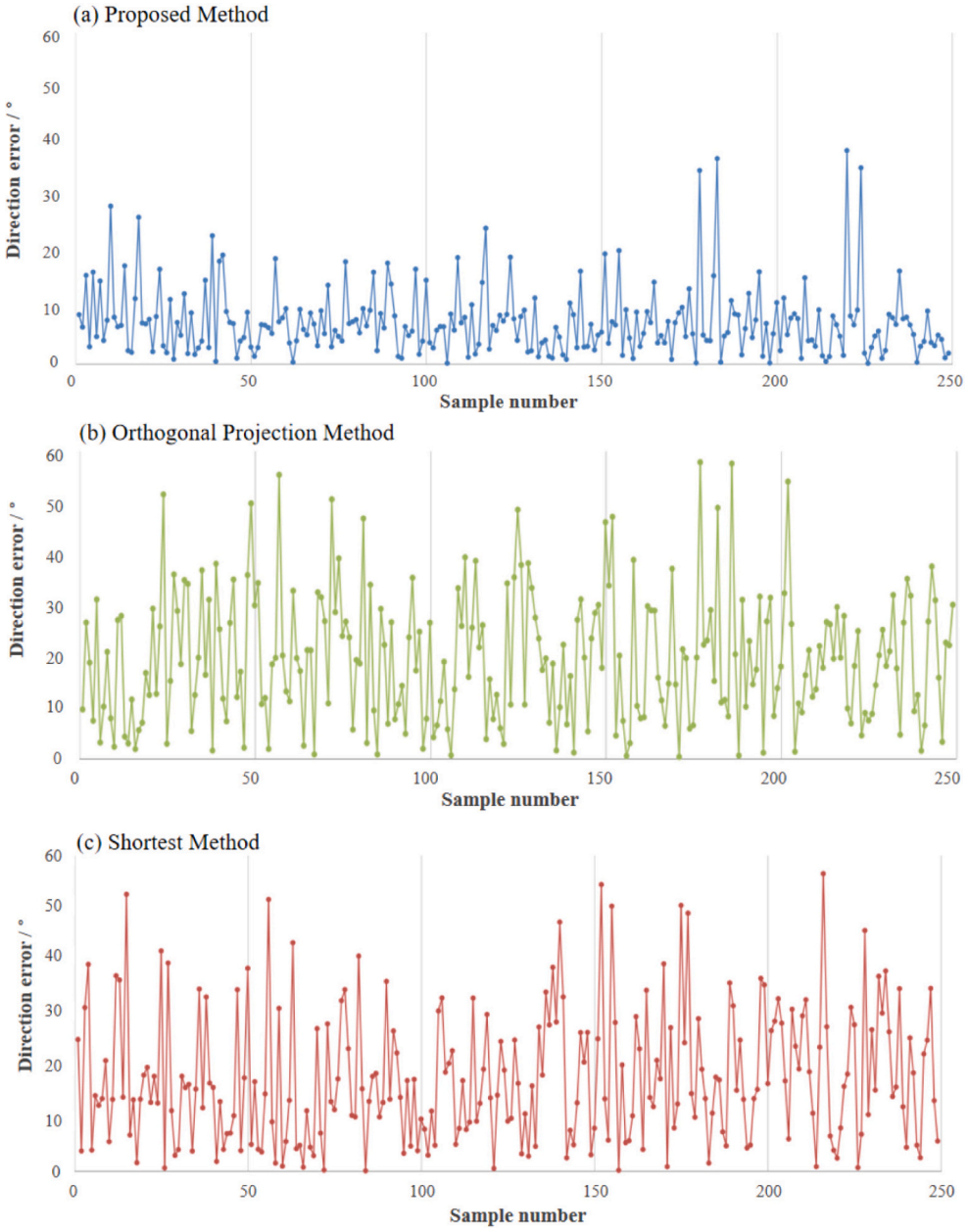


Fig. 15. Distribution of direction errors of width measurement direction determination methods.

four-point configuration. Compared with the 0° pose, the overall distance readings at 45° were larger, consistent with the geometric effect of oblique incidence. Using the 0.8 mm reference strip as a benchmark, the measured widths remained highly consistent across different distances and poses: at 0°, the results were 0.80, 0.81, and 0.77 mm, while at 45°, they were 0.79, 0.83, and 0.76 mm. The corresponding absolute errors ranged from 0 to 0.04 mm, with a mean absolute error of approximately 0.02 mm and an equivalent relative error of about 2.5%, showing no directional bias. The largest deviation occurred under the 6 m, 45° condition, mainly due to reduced pixel resolution at longer distances and slightly lower boundary contrast under oblique imaging, yet the error remained within the acceptable millimeter-level range.

Fig. 17 compares the measured and reference widths, where the horizontal axis represents the reference width and the vertical axis represents the measured width. The left plot corresponds to frontal imaging (0°), with the dashed line $y = x$ indicating ideal consistency and the shaded band representing a ± 4% tolerance. Different distances are marked by circles, squares, and triangles. Most samples align closely with the $y = x$ line and fall within the ±4% band, exhibiting a linear, monotonic relationship without systematic over- or underestimation, indicating a near-unit slope and negligible intercept. The right plot shows oblique imaging (45°) with a relaxed tolerance band of ±6%. Samples at all distances similarly follow the $y = x$ trend within the tolerance range, with slightly increased dispersion for larger widths, reflecting minor effects of reduced contrast and accumulated pixel quantization error at longer viewing distances. Together, the two plots

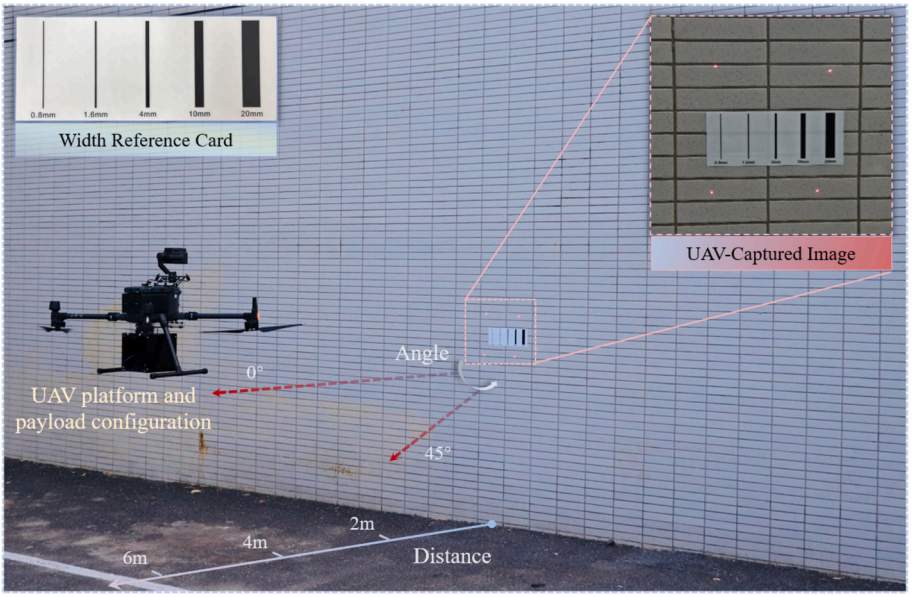
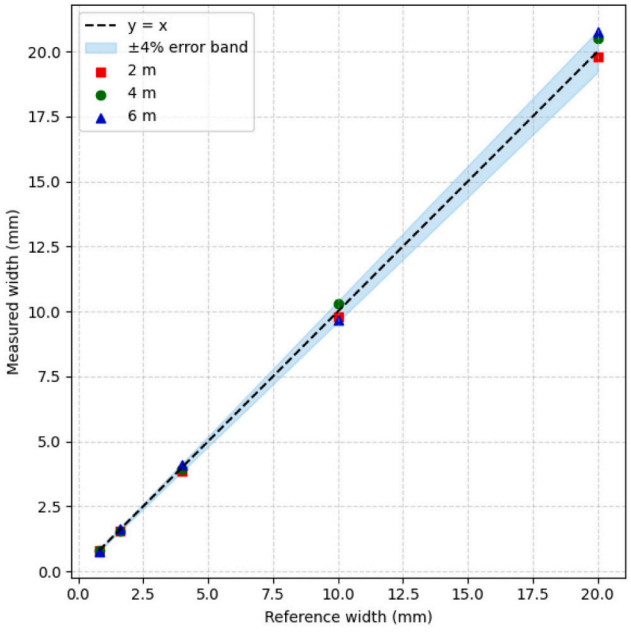


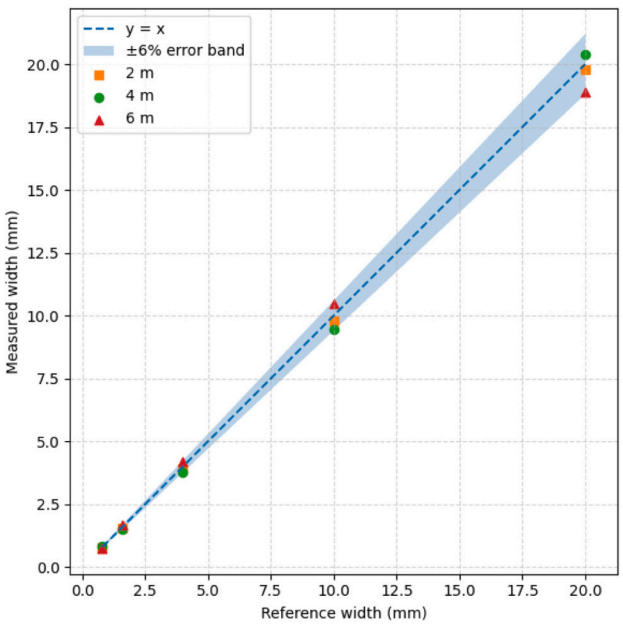
Fig. 16. Field arrangement for pixel-scale validation using a width reference card.

Table 5
Line-width measurement under varying standoff distances and incidence angles.

Distance (m)	Angle (°)	Laser range #1 (mm)	Laser range #2 (mm)	Laser range #3 (mm)	Laser range #4 (mm)	Measured width (mm)	Reference width (mm)
2	0	2001	2000	2000	1999	0.8	0.8
	45	2344	2289	2291	2340	0.79	0.8
4	0	4002	3999	4001	4000	0.81	0.8
	45	4320	4271	4274	4317	0.83	0.8
6	0	6000	6001	5998	6000	0.77	0.8
	45	6378	6327	6324	6379	0.76	0.8



(a) 0° view (±4% error band).



(b) 45° view (±6% error band).

Fig. 17. Width-board validation: agreement between measured and reference widths under varying viewing angles and distances.

demonstrate that the proposed scale calibration framework maintains stable linear consistency and low bias under both frontal and oblique imaging geometries. No observable systematic drift appears across viewing distances, indicating strong invariance and robustness of the

local scale mapping. Overall, the high agreement and low dispersion between measured and reference widths verify the accuracy and stability of the proposed framework for recovering physical pixel scales under multi-view and multi-distance conditions.

6.2. Concrete bridges test

This study selected the Jiyin Bridge in Nanjing, Jiangsu Province, China, as the test subject. The bridge spans the Qinhuai River and is a double-flange continuous beam bridge with a uniform cross-section. It has a total length of approximately 460 m, consisting of a 220 m main span and two 120 m approach spans, with the girder height maintained at about 5 m throughout the structure. As a continuous river-crossing structure with heavy traffic on the deck and an open waterway below affected by strong wind and reflections, the site provides representative field conditions for comprehensively evaluating the UAV-mounted system's attitude stability and measurement loop performance. The test focused on the pier webs and box girder soffits, where structural deterioration commonly occurs. The platform used a DJI M300 RTK multi-rotor UAV equipped with a Zenmuse H20 camera, featuring a 1/1.7-in. CMOS sensor, an equivalent focal length range of 31.7–556.2 mm, and a 20-megapixel resolution, balancing image clarity and field coverage. A self-developed four-point laser ranging device was also mounted as an external absolute scale reference, providing a 0.03-80 m measurement range, ±1 mm absolute accuracy, and a 10-20 Hz working frequency. The experiment was conducted beneath the bridge, where the UAV hovered near the pier webs and box girder soffits for close-range measurements. The payload configuration and structural layout are shown in Fig. 18.

Data acquisition was conducted on designated inspection areas of the pier exterior surface, as shown in Fig. 19. Before flight, the Haiwell touch panel was used to drive the stepper motor according to a preset angular sequence, rotating the four laser ranging sensors to their target orientations. The projected laser spots formed a rectangular array with a defined side length and geometric spread on the pier surface, establishing a traceable initial scale and fixing the corresponding angular parameters. The UAV followed a low-speed lateral flight path close to the pier surface, while RTK positioning provided centimeter-level navigation to ensure consistent trajectory reproduction and spatial reference alignment. During measurement, the four laser channels output distance data at 20 Hz (configurable to 10 Hz depending on task requirements to align with the camera's frame rate and trigger timing). The Haiwell Cloud App enabled remote fine-tuning of the stepper motor from a mobile device to ensure that the four red laser spots were evenly distributed and adequately spread across each image frame. The controller synchronized all distance readings and motor angles with timestamps and locked the camera exposure and ranging sampling through hardware triggering, ensuring strict correspondence between each image frame and its simultaneous four-point geometry. As shown in the lower right of Fig. 19, a typical captured frame contains four evenly distributed red laser spots around the crack area, providing the absolute scale input for that moment. These data directly support the pixel-to-physical distance conversion and the robust estimation of the homography matrix.

Leveraging an integrated software-hardware collaborative data collection process, the system ensures that each image frame has traceable absolute scale and verifiable geometric registration, providing a solid data foundation for subsequent homography estimation and width quantification. Based on this framework, field data collection was conducted on the exterior faces of the piers and the bottom of the box girders. To comprehensively evaluate the robustness of the method in a single bridge scenario, four main cracks (Crack 1 to Crack 4, as shown in Fig. 19 left) with significant morphological differences were specifically selected as test subjects, three representative measurement points (e.g., 1–1, 1–2, 2–1, 2–3, 3–2, 3–3, 4–1, 4–2) were selected on each crack as quantification locations. These samples are highly representative,



Fig. 18. Schematic diagram of bridge overview.



Fig. 19. On-site crack inspection and data acquisition at Jiyin Bridge.

covering various orientations such as longitudinal extension and lateral bending, with crack widths ranging from 0.35 mm for fine cracks to 1.30 mm for more prominent cracks. They also include complex background conditions such as water stain erosion and rough concrete textures. This diverse combination of samples effectively simulates common engineering inspection conditions, ensuring the reliability of the validation results.

To obtain high-confidence ground truth for crack width and validate the UAV measurement results, a portable digital crack width gauge (model CK-102, with a resolution of 0.01 mm) was used for contact measurements at selected test points. To minimize human reading errors and ensure the quality of the benchmark data, a strict repeatability validation protocol was implemented. The same experienced technician measured each test point three times, with the arithmetic mean of the three measurements used as the final result. Statistical analysis showed that the maximum deviation among the three readings was within 0.02 mm, indicating high stability of the measurement process and ensuring that the manually measured data used as a reference in subsequent analyses is metrologically reliable.

The data processing workflow consists of three interconnected stages, namely scale calibration, semantic segmentation, and width quantification, which are sequentially integrated to ensure both geometric and metrological consistency in the results. In the scale calibration stage, based on the four laser distance observations and their known inter-beam angles within the same frame, a four-point reference geometry with absolute physical units is reconstructed on the measured plane following the method described in Section 4.2. The corresponding image containing the four laser spots is processed using zero-mean normalized cross-correlation (ZNCC) within a fixed region of interest to robustly extract the pixel coordinates of the red laser spots. These pixel coordinates are matched one-to-one with the physical reference points reconstructed from distance and angular measurements. According to Section 4.3, the normalized direct linear transform (DLT) is applied to estimate the homography matrix H, establishing the rational mapping from pixel coordinates $(u, v)$to physical plane coordinates $(X(u, v), Y(u, v))$. Partial derivatives of $(X(u, v), Y(u, v))$ with respect to (u, v) yield the Jacobian matrix: $J(u, v) = \frac{\partial(X,Y)}{\partial(u,v)} = \begin{bmatrix} \frac{\partial X}{\partial u} & \frac{\partial X}{\partial v} \\ \frac{\partial Y}{\partial u} & \frac{\partial Y}{\partial v} \end{bmatrix}$. This enables the definition of a local length metric $\| J(u, v)\mathbf{t}\|_2$ (unit: mm·px^{-1}) at any position and direction $\mathbf{t}$, providing a geometry-consistent, position-dependent scale without resampling the entire image. In the semantic segmentation stage, the original image is processed using CrackMamba-Net to generate a high-precision crack mask (see Fig. 20). The network, developed on the foundation of state space modeling, integrates global dependency and local texture representation, ensuring continuity of crack centerlines and clear boundary delineation under weak contrast and fine crack conditions. In the width quantification stage, the segmented mask is refined through topology-

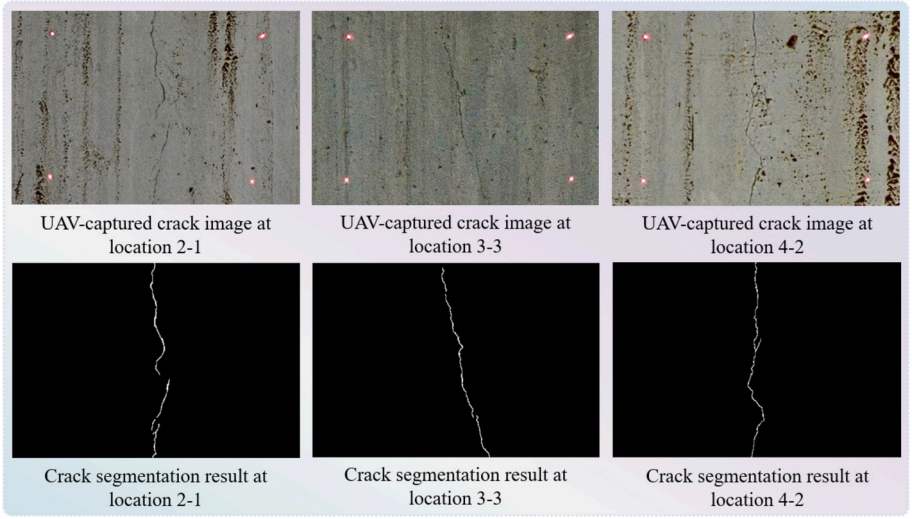


Fig. 20. UAV-captured crack images and corresponding segmentation results.

preserving thinning to obtain a single-pixel skeleton, followed by skeleton purification using length thresholds and spur suppression. For each local neighborhood, principal component analysis (PCA) is applied to estimate the dominant skeleton direction and derive the initial normal. To mitigate orientation jitter caused by pixelation, boundary points along the main axis are fitted using a distance-weighted first-order least squares model, achieving local linear refinement. Finally, rays are projected or intersected along the refined normal on both sides of the crack boundary to compute the pixel-domain normal width, which is then converted to physical units using the previously derived local length metric.

Table 6 provides a detailed comparison of the quantification results for 8 representative measurement points across the 4 main cracks of the Jiyin Bridge, including the manual reference values, algorithmic estimates, absolute errors, and relative accuracy. To rigorously assess the statistical reliability of the measurement results, the 95% confidence interval (95% CI) of the error distribution was also calculated. The quantitative analysis reveals that the mean absolute error is 0.038 mm, with a 95% confidence interval of [0.029, 0.047] mm, indicating high precision and stability in the measurements. The mean signed error is −0.0025 mm, with the corresponding 95% confidence interval of [−0.036, 0.031] mm. This interval includes zero, statistically confirming that the proposed method does not exhibit significant systematic overestimation or underestimation. The accuracy of individual measurement points ranges from 91.4% to 97.3%, all satisfying the engineering precision requirement at the millimeter scale. Wider cracks, such as points 2–1 (1.12 mm) and 4–1 (1.26 mm), exhibit higher accuracies between 96.8% and 97.3%, while finer submillimeter to 0.5 mm cracks (e.g., 1–2, 2–3, 3–2, and 4–2) show slightly higher relative errors, with accuracies between 91.4% and 94.5%. This observation aligns with the expected influence of pixelization and boundary contrast, which are more sensitive to fine-scale measurements. When grouped by crack ID, the error levels remain comparable within each crack, demonstrating good transferability and stability of the scale calibration under varying viewing angles, texture backgrounds, and curvature conditions. The largest observed deviation, such as −0.06 mm at point 4–2, is primarily attributed to local surface texture, uneven illumination, or boundary blurring, yet the magnitude remains well within acceptable engineering tolerances.

To further evaluate the stability of the measurement system across continuous space, this study implemented a high-density sampling strategy, evenly distributing 10 discrete measurement points (P1–P10) along the extension path of each crack. In Fig. 21, the blue curves represent manually measured reference widths, while the orange curves show the corresponding estimates obtained by the proposed method. The two curves exhibit highly consistent trends across all four cracks, with well-matched peak positions, aligned width fluctuations, and no significant phase deviations or cumulative systematic drift along the measurement path. Overall, the field results confirm the robustness and metrological consistency of the proposed framework under complex on-site conditions. The method achieves millimeter-level agreement with manual measurements and maintains low bias and strong repeatability across different cracks, positions, and viewing angles, thereby providing a reliable quantitative basis for the assessment of bridge surface defects.

Table 6
Width calculation results of Jiyin Bridge tests.

Crack	Crack location	Manual measurement (mm)	Proposed method (mm)	Absolute error(mm)	Accuracy (%)
1–1	**Crack 1**	**0.77**	**0.74**	**−0.03**	**96.10**
1–2		0.48	0.52	0.04	91.67
2–1	Crack 2	1.12	1.09	−0.03	97.32
2–3		0.35	0.38	0.03	91.43
3–2	Crack 3	0.62	0.58	−0.04	93.55
3–3		0.54	0.57	0.03	94.44
4–1	Crack 4	1.26	1.30	0.04	96.83
4–2		0.89	0.83	−0.06	93.26

7. Conclusions

This paper addressed the core challenge of achieving geometrically accurate and millimeter-scale quantification of crack width under UAV-based oblique imaging and variable distance conditions. A complete technical framework is proposed and systematically validated, integrating external absolute scale injection, visual segmentation, and geometric measurement into a closed-loop process.

At the methodological level, a four-point laser ranging device mounted on the UAV serves as the external scale source. Within each synchronized frame, the measured distances and inter-beam angles are used to reconstruct a four-point reference geometry with absolute physical units on the target plane. On the image plane, the pixel coordinates of the four laser spots are robustly extracted using fixed-region zero-mean normalized cross-correlation matching. The normalized direct linear transform is then applied to establish the homography between pixel and physical planes. By differentiating the rational mapping, the Jacobian matrix is obtained, allowing the computation of position- and direction-dependent local length metrics. Consequently, the scale factor is transformed from a global constant into a metric function adaptive to local geometry and imaging configuration, effectively suppressing scale drift caused by oblique views and variable distances.

For visual recognition, a crack segmentation network, termed CrackMamba-Net, is developed based on state space modeling. The network integrates global-local vision state space modules with attention and multi-scale fusion mechanisms, balancing long-range dependency and boundary detail representation to produce high-precision crack masks. Comparative experiments demonstrate that CrackMamba-Net achieves superior continuity of crack orientation and sharper boundary delineation under weak contrast, fine cracks, and complex backgrounds. The network outperforms seven representative baselines in key metrics including mIoU, precision, recall, and F1, while maintaining a deployable level of parameter size and computational cost, providing a stable segmentation prior for subsequent geometric measurement.

In the geometric measurement stage, this study strictly follows the engineering definition of "the shortest physical distance between two crack boundaries along the local normal direction." The crack skeleton is first refined to ensure topological reliability. The local dominant orientation is then estimated using principal component analysis (PCA), followed by orientation refinement through a distance-weighted linear model. Finally, intersections between the refined normal and both crack boundaries are determined, and the physical distance between these intersections is calculated as the crack width. Experimental results based on manually annotated reference directions show that the mean and median orientation errors are 6.74° and 6.56°, respectively, with a standard deviation of 5.41°. These results are significantly lower than those obtained using the orthogonal projection and shortest-distance methods, demonstrating superior consistency and robustness across varying texture, width, and curvature conditions.

Field and in situ bridge experiments further validate the effectiveness and transferability of the proposed scale calibration and width measurement framework. In the calibration test using a dimensional reference board, the local length metric exhibited strong linear consistency with the physical scale under three working distances (2, 4, and 6 m) and two imaging poses (frontal and 45° oblique). The mean absolute error for the 0.8 mm stripe was approximately 0.02 mm, with no observable systematic bias. In the Jiyin Bridge field test, the average absolute error for eight representative points across four cracks was about 0.038 mm, with a signed mean of −0.0025 mm and an accuracy range of 91.4%–97.3%. The system maintained millimeter-scale quantification capability under challenging conditions such as complex lighting, surface

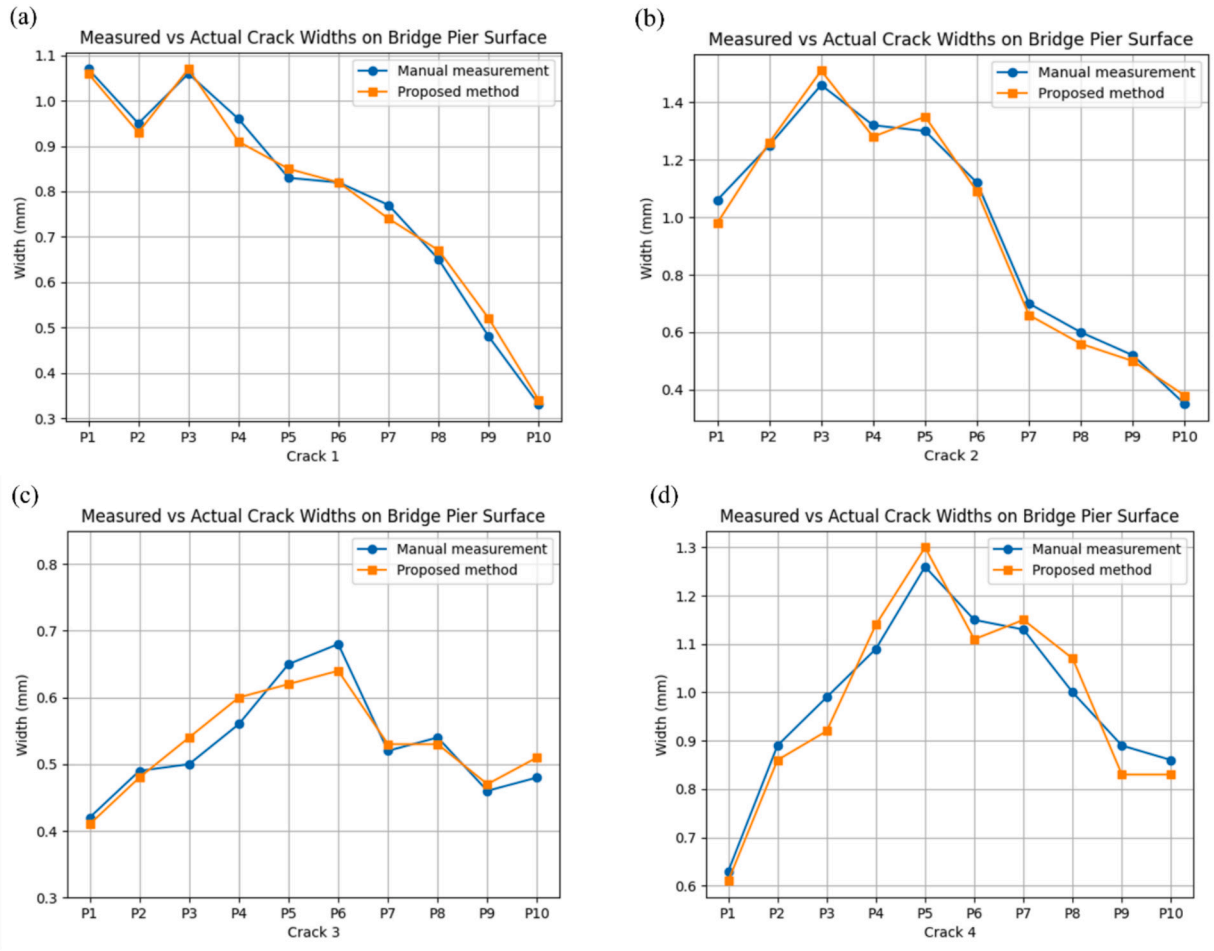


Fig. 21. Comparison of measured and actual crack widths for four cracks in the field test: (a) Crack 1, (b) Crack 2, (c) Crack 3, and (d) Crack 4.

reflections, and wind disturbances, demonstrating robustness against variations in distance, pose, and background.

Although this study demonstrates the effectiveness of UAV-based crack quantification, several limitations remain. First, the ZNCC matching and red-channel enhancement strategy can fail under direct sunlight, strong reflections, or extremely low contrast, where the signal-to-noise ratio of the laser spots drops. Future work will consider high-power modulated lasers, narrow-band optical filtering, and adaptive exposure control to improve ranging stability under challenging illumination. Second, the scale recovery relies on a local planar assumption. For curved components such as circular piers or highly uneven surfaces, homography-based projection may introduce geometric distortions. Integrating sparse depth constraints or Structure-from-Motion reconstruction will be explored to extend applicability to non-planar geometries. Third, although field tests show millimeter-level accuracy, a complete analytical error propagation model has yet to be established, and residual synchronization errors during rapid maneuvers may still affect scale estimation. Future efforts will develop a rigorous theoretical error propagation framework supported by Monte Carlo simulations, and incorporate motion-compensation or hardware-level synchronization methods to reduce dynamic spatiotemporal deviations. Finally, data collection and validation were conducted on a single bridge. Broader cross-region and cross-structure evaluation is required, and future work will construct a multi-source heterogeneous crack dataset to improve generalization across diverse bridge types and material conditions.

In summary, the proposed closed-loop framework, integrating scale calibration and normal-based quantification, provides a reusable technical pathway and empirical foundation for millimeter-scale, traceable crack measurement in UAV-based applications.

CRediT authorship contribution statement

Jinghuan Zhang: Writing – original draft, Software, Methodology, Data curation. **Wang Chen:** Writing – review & editing, Software. **Jian Zhang:** Supervision, Project administration, Investigation, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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